

CSE111: Orientation to Computing

Sample Question Papers & University Examination Suite

Complete 16-Paper Assessment Suite (Sets A, B, C & D)

All Papers 100% Objective MCQ Format with Detailed Solutions

4 Continuous Assessment 1 (CA-1) Papers	Units 1 & 2
4 Mid-Term Examination Papers	Units 1, 2 & 3
4 Continuous Assessment 2 (CA-2) Papers	Units 4 & 5
4 End-Term Examination Papers	All Units (1 to 6)

Standard University Exam Level: 60% Direct Recall, 25–30% Conceptual, 10% Application

Prepared for Lovely Professional University (LPU) Students

Strictly Calibrated to University Assessment Patterns & Previous Year Questions

LPU Engineering Open Series

Academic Session 2025–2026

Examination Structure & Blueprint

Overview of Assessment Components

This volume contains the complete official evaluation suite for **CSE111: Orientation to Computing**, organized into 16 distinct model test papers across four major academic milestones:

I. Continuous Assessment 1 (CA-1) Suite [Sets A, B, C, D]:

- **Syllabus Scope:** Unit 1 (Computer Languages) and Unit 2 (Computer Fundamentals).
- **Question Format:** 25 Multiple-Choice Questions (1 mark each). Total = 25 Marks.
- **Time Allowed:** 45 Minutes.

II. Mid-Term Examination Suite [Sets A, B, C, D]:

- **Syllabus Scope:** Unit 1 (Computer Languages), Unit 2 (Computer Fundamentals), and Unit 3 (Computer Hardware).
- **Question Format:** 30 Multiple-Choice Questions (1 mark each). Total = 30 Marks.
- **Time Allowed:** 60 Minutes.

III. Continuous Assessment 2 (CA-2) Suite [Sets A, B, C, D]:

- **Syllabus Scope:** Unit 4 (Number Systems) and Unit 5 (Version Control).
- **Question Format:** 25 Multiple-Choice Questions (1 mark each). Total = 25 Marks.
- **Time Allowed:** 45 Minutes.

IV. End-Term Examination Suite [Sets A, B, C, D]:

- **Syllabus Scope:** Comprehensive coverage across all six units (Units 1 to 6).
- **Question Format:** 30 Multiple-Choice Questions (1 mark each). Total = 30 Marks.
- **Time Allowed:** 90 Minutes.

Difficulty Calibration Standard

In direct alignment with the university previous-year question pattern:

- **60% Direct Recall / Definitions:** Testing core terminology, acronyms, and direct facts.
- **25–30% Basic Conceptual Understanding:** Understanding system behavior, purpose of hardware units, and Git commands.
- **10% Simple Application:** Standard base conversions (binary, octal, hex) and simple tracing.
- **0% Hard / Competitive Traps:** No multi-step calculations, deep architecture proofs, or confusing double negatives.

Contents

Examination Structure & Blueprint	i
1 Continuous Assessment 1 (CA-1) Suite	1
1.1 Continuous Assessment 1 — Set A	2
1.2 Continuous Assessment 1 — Set B	8
1.3 Continuous Assessment 1 — Set C	14
1.4 Continuous Assessment 1 — Set D	20
2 Mid-Term Examination Suite	26
2.1 Mid-Term Examination — Set A	27
2.2 Mid-Term Examination — Set B	34
2.3 Mid-Term Examination — Set C	41
2.4 Mid-Term Examination — Set D	48
3 Continuous Assessment 2 (CA-2) Suite	55
3.1 Continuous Assessment 2 — Set A	56
3.2 Continuous Assessment 2 — Set B	62
3.3 Continuous Assessment 2 — Set C	68
3.4 Continuous Assessment 2 — Set D	74
4 End-Term Examination Suite (Comprehensive)	80
4.1 End-Term Examination — Set A	81
4.2 End-Term Examination — Set B	88
4.3 End-Term Examination — Set C	95
4.4 End-Term Examination — Set D	102

Chapter 1

Continuous Assessment 1 (CA-1) Suite

Continuous Assessment 1 evaluates students on **Unit 1 (Computer Languages)** and **Unit 2 (Computer Fundamentals)**. Each set contains **25 Multiple-Choice Questions (1 mark each, Total = 25 Marks, Time Allowed = 45 Minutes)**. Complete Answer Keys and detailed explanations are provided after each paper.

1.1 Continuous Assessment 1 — Set A

LOVELY PROFESSIONAL UNIVERSITY

Department of Computer Science & Engineering

Continuous Assessment 1 — SET A

Course Code: CSE111
Max. Marks: 25 Marks

Title: Orientation to Computing
Syllabus Scope: Unit 1 (Computer Languages) &
Unit 2 (Computer Fundamentals)

Time Allowed: 45 Minutes
Format: 100% MCQ

Instructions: All 25 questions are compulsory. Each question carries 1 mark. Select the single best answer.

Q.1 Which of the following languages is executed directly by the CPU without translation?

- (A) Assembly Language
- (B) Machine Language
- (C) Python
- (D) C++

Q.2 In assembly language, readable abbreviations like ADD and MOV are known as:

- (A) Operands
- (B) Mnemonics
- (C) Macros
- (D) Pointers

Q.3 A software tool that converts high-level source code into machine code statement-by-statement at runtime is a/an:

- (A) Compiler
- (B) Interpreter
- (C) Assembler
- (D) Linker

Q.4 Which phase of compilation converts character streams into tokens like identifiers and keywords?

- (A) Syntax Analysis
- (B) Lexical Analysis
- (C) Code Optimization
- (D) Code Generation

Q.5 What is the primary role of the Linker in program translation?

- (A) To edit source code
- (B) To combine object files and libraries into an executable binary
- (C) To load instructions into RAM
- (D) To debug runtime logic

Q.6 Which operating system utility loads the compiled executable file into primary memory (RAM) for execution?

- (A) Compiler
- (B) Loader
- (C) Assembler
- (D) Text Editor

Q.7 What does an assembler do in its first pass?

- (A) Generates final machine instructions
- (B) Records symbols and label definitions with their memory addresses
- (C) Links external libraries
- (D) Executes instructions

Q.8 Which type of programming error allows a program to compile and run, but yields unintended or incorrect results?

- (A) Syntax error
- (B) Logic error
- (C) Compile-time error
- (D) Linker error

Q.9 What is the correct sequence of steps in the CPU instruction cycle?

- (A) Decode → Execute → Fetch → Store
- (B) Fetch → Decode → Execute → Store
- (C) Store → Fetch → Execute → Decode
- (D) Execute → Decode → Fetch → Store

Q.10 A finite, ordered set of step-by-step instructions to solve a given computing problem is called a/an:

- (A) Mnemonic
- (B) Algorithm
- (C) Operand
- (D) Token

Q.11 In the C/C++ compilation pipeline, which tool expands `#include` directives and macros?

- (A) Linker
- (B) Preprocessor
- (C) Loader
- (D) Assembler

Q.12 The architecture where program instructions and data share the same internal memory was proposed in 1945 by:

- (A) Charles Babbage
- (B) John von Neumann
- (C) Blaise Pascal
- (D) Alan Turing

Q.13 What electronic switching component was used in first-generation computers (1940–1956)?

- (A) Transistors
- (B) Vacuum tubes
- (C) Integrated circuits
- (D) Microprocessors

Q.14 Second-generation computers (1956–1963) replaced vacuum tubes with:

- (A) Microprocessors
- (B) Transistors
- (C) Integrated circuits
- (D) Magnetic drums

Q.15 Which generation of computers first used Integrated Circuits (ICs) on silicon chips?

- (A) First generation
- (B) Second generation
- (C) Third generation
- (D) Fourth generation

Q.16 The first commercial single-chip microprocessor (Intel 4004) was launched in:

- (A) 1956
- (B) 1964
- (C) 1971
- (D) 1981

Q.17 Which CPU component performs arithmetic additions and logical comparisons?

- (A) Control Unit
- (B) Arithmetic Logic Unit (ALU)
- (C) Program Counter
- (D) Address Bus

Q.18 Which CPU component coordinates operations by generating timing and control signals?

- (A) Control Unit (CU)
- (B) ALU
- (C) RAM
- (D) Hard Disk

Q.19 Which system bus is unidirectional and transmits memory addresses from the CPU to memory?

- (A) Data Bus
- (B) Address Bus
- (C) Control Bus
- (D) Serial Bus

Q.20 Which system bus carries the actual data and machine instructions between hardware units?

- (A) Control Bus
- (B) Data Bus
- (C) Address Bus
- (D) Power Bus

Q.21 The computer principle stating that inaccurate input data produces inaccurate output information is:

- (A) FIFO
- (B) GIGO (Garbage In, Garbage Out)
- (C) LIFO
- (D) DMA

Q.22 Which type of computer is designed for computation-intensive scientific tasks like climate modeling?

- (A) Microcontroller
- (B) Supercomputer
- (C) Workstation
- (D) Embedded system

Q.23 Large, highly reliable computers used by banks and airlines for massive transaction processing are:

- (A) Personal computers
- (B) Mainframe computers
- (C) Workstations
- (D) Laptops

Q.24 A small computer built into a washing machine or automobile engine to perform a single dedicated control function is a/an:

- (A) Supercomputer
- (B) Embedded computer
- (C) Mainframe
- (D) Server

Q.25 A computer that processes continuous physical variables like electrical voltage rather than discrete binary digits is a/an:

- (A) Digital computer
- (B) Analog computer
- (C) Hybrid computer
- (D) Mainframe

Answer Key & Explanations — CA-1 (Set A)

CA-1 Set A — Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	6	B	11	B	16	C	21	B
2	B	7	B	12	B	17	B	22	B
3	B	8	B	13	B	18	A	23	B
4	B	9	B	14	B	19	B	24	B
5	B	10	B	15	C	20	B	25	B

Technical Solutions — CA-1 Set A

- **Q.1 (B):** Machine language is the only language a processor's digital circuits execute natively.
- **Q.3 (B):** Interpreters evaluate and execute code line-by-line during runtime, unlike compilers which translate everything beforehand.
- **Q.5 (B):** The linker links external library code (e.g., `printf`) and object files to create a standalone executable.
- **Q.12 (B):** John von Neumann introduced the stored-program architecture in 1945.
- **Q.16 (C):** The Intel 4004 microprocessor was introduced in 1971, marking the beginning of 4th-generation computing.
- **Q.19 (B):** The Address Bus is unidirectional; memory addresses originate only from the CPU.
- **Q.24 (B):** Embedded computers are dedicated microcontrollers engineered into non-computer devices for specialized control.

1.2 Continuous Assessment 1 — Set B

LOVELY PROFESSIONAL UNIVERSITY

Department of Computer Science & Engineering

Continuous Assessment 1 — SET B

Course Code: CSE111

Title: Orientation to Computing

Time Allowed: 45 Minutes

Max. Marks: 25 Marks

Syllabus Scope: Unit 1 (Computer Languages) &
Unit 2 (Computer Fundamentals)

Format: 100% MCQ

Instructions: All 25 questions are compulsory. Each question carries 1 mark. Select the single best answer.

Q.1 Which of the following is a key characteristic of high-level programming languages?

- (A) They use only binary opcodes
- (B) They provide hardware abstraction and human-readable syntax
- (C) They cannot be translated
- (D) They run only on one processor architecture

Q.2 A translator that produces an independent executable object file before running the program is a/an:

- (A) Interpreter
- (B) Compiler
- (C) Debugger
- (D) Text Editor

Q.3 In assembly language instruction `MOV R1, R2`, what are R1 and R2?

- (A) Mnemonics
- (B) CPU Registers
- (C) Immediate constants
- (D) Labels

Q.4 Which compilation phase checks whether expressions follow type compatibility rules?

- (A) Lexical Analysis
- (B) Syntax Analysis
- (C) Semantic Analysis
- (D) Preprocessing

Q.5 Division by zero while running a program is an example of a:

- (A) Syntax error
- (B) Logic error
- (C) Runtime error
- (D) Linker error

Q.6 What is the primary function of an Assembler?

- (A) Translating high-level language to assembly
- (B) Translating assembly language mnemonics into binary machine code
- (C) Running programs line-by-line
- (D) Connecting to the internet

Q.7 Which step in the program development cycle involves using test cases to find and fix bugs?

- (A) Coding
- (B) Testing and Debugging
- (C) Problem Definition
- (D) Documentation

Q.8 A graphical chart that depicts program logic using diamonds, rectangles, and arrows is a:

- (A) Pseudocode
- (B) Flowchart
- (C) Syntax diagram
- (D) Data sheet

Q.9 In an assembler, what occurs during the second pass?

- (A) Recording label addresses
- (B) Generating machine instructions and resolving address references
- (C) Lexical scanning only
- (D) Loading the executable

Q.10 Which language provides direct register-level control while remaining symbolic?

- (A) Machine Language
- (B) Assembly Language
- (C) Python
- (D) HTML

Q.11 Who invented the Pascaline calculating machine in 1642?

- (A) Charles Babbage

- (B) Blaise Pascal
- (C) Herman Hollerith
- (D) Alan Turing

Q.12 The concept of storing both program instructions and working data in the same internal memory is the:

- (A) Babbage concept
- (B) Stored-program principle
- (C) GIGO principle
- (D) RISC principle

Q.13 Which hardware technology was used as the primary internal logic switch in first-generation computers?

- (A) Transistors
- (B) Vacuum tubes
- (C) Integrated circuits
- (D) Microprocessors

Q.14 Which high-level programming languages were widely popularized during the second computer generation?

- (A) Python and Java
- (B) FORTRAN and COBOL
- (C) HTML and SQL
- (D) C# and Swift

Q.15 The third generation of computing was defined by the introduction of:

- (A) Vacuum tubes
- (B) Discrete relays
- (C) Integrated Circuits (ICs)
- (D) Microcontrollers

Q.16 The Intel 4004, launched in 1971, is celebrated in computing history as the world's first:

- (A) Supercomputer
- (B) Single-chip microprocessor
- (C) Hard disk drive
- (D) Operating system

Q.17 Fifth-generation computing primarily aims to advance systems in:

- (A) Vacuum tube amplification
- (B) Mechanical levers
- (C) Artificial intelligence and parallel computing
- (D) Punched card sorting

Q.18 The computer characteristic that allows it to execute millions of calculations without fatigue is:

- (A) Versatility
- (B) Accuracy
- (C) Diligence
- (D) Automation

Q.19 Which sub-unit of the CPU performs logic operations like AND, OR, and comparisons?

- (A) Control Unit
- (B) Arithmetic Logic Unit (ALU)
- (C) Instruction Register
- (D) Address Bus

Q.20 Which system bus transmits signals such as Memory Read, Memory Write, and Clock Timing?

- (A) Data Bus
- (B) Address Bus
- (C) Control Bus
- (D) Serial Bus

Q.21 Which of the following is volatile primary memory?

- (A) ROM
- (B) Hard Disk
- (C) RAM
- (D) Flash Drive

Q.22 Firmware that initializes computer hardware during startup (BIOS/UEFI) is stored in:

- (A) RAM
- (B) ROM
- (C) Cache
- (D) Secondary Storage

Q.23 Software that directly manages computer hardware and provides a platform for user applications is:

- (A) Application Software
- (B) System Software
- (C) Shareware
- (D) Freeware

Q.24 Computers designed to process discrete binary digits (0 and 1) are classified as:

- (A) Analog computers
- (B) Digital computers
- (C) Hybrid computers
- (D) Mechanical engines

Q.25 A server computer's primary role in a network is to:

- (A) Act as a game controller
- (B) Provide shared resources, services, and data to client computers
- (C) Operate washing machine motors
- (D) Eliminate RAM

Answer Key & Explanations — CA-1 (Set B)

CA-1 Set B — Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	6	B	11	B	16	B	21	C
2	B	7	B	12	B	17	C	22	B
3	B	8	B	13	B	18	C	23	B
4	C	9	B	14	B	19	B	24	B
5	C	10	B	15	C	20	C	25	B

Technical Solutions — CA-1 Set B

- **Q.2 (B):** Compilers translate entire source files to machine/object code, allowing standalone execution without re-translation.
- **Q.4 (C):** Semantic analysis verifies type rules, such as checking if operands match operators.
- **Q.9 (B):** In pass 2, an assembler translates instructions into machine code using the symbol table created in pass 1.
- **Q.12 (B):** Stored-program principle allows programs to be loaded into RAM dynamically just like data.
- **Q.18 (C):** Diligence is the ability of computers to perform monotonous, repetitive tasks without fatigue.
- **Q.21 (C):** RAM loses all stored data when power is disconnected (volatile).

1.3 Continuous Assessment 1 — Set C

LOVELY PROFESSIONAL UNIVERSITY

Department of Computer Science & Engineering

Continuous Assessment 1 — SET C

Course Code: CSE111
Max. Marks: 25 Marks

Title: Orientation to Computing
Syllabus Scope: Unit 1 (Computer Languages) &
Unit 2 (Computer Fundamentals)

Time Allowed: 45 Minutes
Format: 100% MCQ

Instructions: All 25 questions are compulsory. Each question carries 1 mark. Select the single best answer.

Q.1 Which programming language does not require any translator before direct execution by the processor?

- (A) Assembly Language
- (B) Machine Language
- (C) C
- (D) Java

Q.2 What is the binary representation of instructions in machine language composed of?

- (A) Bits 0 and 1
- (B) Alphanumeric strings
- (C) XML tags
- (D) English keywords

Q.3 An immediate feedback execution model where statements are evaluated one-by-one is provided by a/an:

- (A) Compiler
- (B) Interpreter
- (C) Assembler
- (D) Linker

Q.4 Which part of the compiler identifies syntax errors like missing semicolons or unmatched parentheses?

- (A) Lexical Analysis
- (B) Syntax Analysis
- (C) Semantic Analysis
- (D) Code Generation

Q.5 Connecting external function definitions from libraries (e.g., standard math library) to an object file is done by the:

- (A) Loader
- (B) Linker
- (C) Preprocessor
- (D) Editor

Q.6 Which system software puts the compiled and linked binary file into main memory to begin execution?

- (A) Linker
- (B) Loader
- (C) Compiler
- (D) Debugger

Q.7 Why are assembly language programs less portable than high-level programs?

- (A) They are too large in size
- (B) They are tied directly to a specific processor's instruction set architecture
- (C) They cannot execute arithmetic
- (D) They lack compilers

Q.8 Testing a program with input value 0 when calculating division is called:

- (A) Syntax checking
- (B) Boundary value testing
- (C) Optimization
- (D) Semantic scanning

Q.9 Which stage of the CPU instruction cycle retrieves the binary instruction from RAM into the CPU?

- (A) Execute
- (B) Fetch
- (C) Decode
- (D) Store

Q.10 Which step of the program development cycle involves ongoing bug fixes and functional enhancements after deployment?

- (A) Analysis
- (B) Maintenance
- (C) Coding

(D) Design

Q.11 Who proposed the mechanical Analytical Engine with basic concepts of input, processing, memory, and output in the 1830s?

(A) Blaise Pascal

(B) Charles Babbage

(C) Herman Hollerith

(D) John von Neumann

Q.12 Herman Hollerith used punched cards to process data for which major event?

(A) 1890 US Census

(B) Apollo 11 mission

(C) 1945 World War II codebreaking

(D) 1971 microprocessor design

Q.13 First-generation computers were plagued by which physical drawback?

(A) Very low power consumption

(B) Excessive heat production and frequent vacuum tube burnouts

(C) Excessive microchip speed

(D) Complete portability

Q.14 Second-generation computers (1956–1963) saw the introduction of:

(A) Vacuum tubes

(B) Transistors

(C) Microprocessors

(D) Artificial Intelligence

Q.15 Which generation of computers introduced user interaction through keyboards and monitors rather than paper punch cards?

(A) First generation

(B) Third generation

(C) Second generation

(D) Fifth generation

Q.16 The fourth generation of computers began around 1971 with the invention of the:

(A) Vacuum tube

(B) Microprocessor

(C) Transistor

(D) Analytical Engine

Q.17 Fifth-generation computers are distinguished by an emphasis on:

(A) Natural language processing and Artificial Intelligence

(B) Relay switches

(C) Punched card tabulators

(D) Analog vacuum tubes

Q.18 The capability of a computer to handle diverse tasks like spreadsheets, video playback, and gaming is:

(A) Diligence

(B) Versatility

(C) Accuracy

(D) Storage

Q.19 Which CPU unit directs data flow between registers, the ALU, and system buses?

(A) Control Unit (CU)

(B) Arithmetic Logic Unit

(C) Heat sink

(D) Storage unit

Q.20 Which system bus is bidirectional and transfers data words between the CPU and memory?

(A) Address Bus

(B) Data Bus

(C) Control Bus

(D) Serial Bus

Q.21 Which memory type provides ultra-fast temporary storage located directly inside the CPU chip?

(A) Registers

(B) Hard drive

(C) Flash drive

(D) Optical disc

Q.22 Why is RAM considered volatile memory?

(A) It catches fire easily

(B) It loses its stored data as soon as power is disconnected

(C) It cannot be overwritten

(D) It is mechanical

Q.23 Which type of software includes web browsers, photo editors, and spreadsheet programs?

(A) System software

(B) Application software

(C) Firmware

(D) Device driver

Q.24 Computers that combine both analog measurement and digital processing capabilities are:

(A) Digital computers

(B) Hybrid computers

(C) Supercomputers

(D) Microcontrollers

Q.25 A computer built directly into an anti-lock braking system (ABS) to perform dedicated real-time control is a/an:

(A) Supercomputer

(B) Embedded computer

(C) Mainframe

(D) Workstation

Answer Key & Explanations — CA-1 (Set C)

CA-1 Set C — Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	6	B	11	B	16	B	21	A
2	A	7	B	12	A	17	A	22	B
3	B	8	B	13	B	18	B	23	B
4	B	9	B	14	B	19	A	24	B
5	B	10	B	15	B	20	B	25	B

Technical Solutions — CA-1 Set C

- **Q.4 (B):** Syntax analysis parses tokens to ensure statements conform to grammatical grammar rules.
- **Q.7 (B):** Assembly language mnemonics map 1-to-1 with the specific hardware opcodes of that processor model.
- **Q.15 (B):** Third-generation computers brought monitors and keyboards, moving away from punched cards.
- **Q.20 (B):** The Data bus is bidirectional because the CPU both reads from and writes to memory and I/O.
- **Q.21 (A):** Internal CPU registers are the closest, fastest storage locations in computing hardware.

1.4 Continuous Assessment 1 — Set D

LOVELY PROFESSIONAL UNIVERSITY

Department of Computer Science & Engineering

Continuous Assessment 1 — SET D

Course Code: CSE111

Title: Orientation to Computing

Time Allowed: 45 Minutes

Max. Marks: 25 Marks

Syllabus Scope: Unit 1 (Computer Languages) &
Unit 2 (Computer Fundamentals)

Format: 100% MCQ

Instructions: All 25 questions are compulsory. Each question carries 1 mark. Select the single best answer.

Q.1 In which language does each instruction typically represent a 1-to-1 correspondence with a machine instruction using mnemonics?

- (A) Assembly Language
- (B) Python
- (C) Java
- (D) SQL

Q.2 Which tool translates an entire high-level program and reports compile-time errors prior to runtime?

- (A) Interpreter
- (B) Compiler
- (C) Operating System
- (D) Text Editor

Q.3 In an assembly instruction `ADD R1, R2, R3`, `R1`, `R2`, and `R3` are:

- (A) Opcodes
- (B) Operands / Registers
- (C) Preprocessors
- (D) Labels

Q.4 The phase in compilation that transforms source code into optimized lower-level instructions is:

- (A) Code generation & optimization
- (B) Lexical analysis
- (C) Preprocessing
- (D) Linking

Q.5 A program containing a logic error will:

- (A) Fail during compilation with a syntax error
- (B) Compile and run, but output wrong results
- (C) Freeze the CPU indefinitely
- (D) Refuse to load into RAM

Q.6 What is the primary role of the operating system's loader?

- (A) Translating high-level source code to assembly
- (B) Placing the executable image into physical RAM to initiate execution
- (C) Compiling Java bytecode
- (D) Formatting hard drives

Q.7 What does the preprocessor in C/C++ handle before compilation begins?

- (A) Executing the program
- (B) Header file inclusions (`#include`) and macro replacements
- (C) Linking object files
- (D) Syntax tree generation

Q.8 Which step of the program development cycle involves formalizing the inputs, outputs, and requirements of the problem?

- (A) Problem Definition
- (B) Coding
- (C) Testing
- (D) Maintenance

Q.9 A step-by-step description of an algorithm using plain English mixed with programming-like structures is:

- (A) Pseudocode
- (B) Machine code
- (C) Bytecode
- (D) Microcode

Q.10 During which phase of a two-pass assembler are memory addresses resolved and machine code emitted?

- (A) First pass
- (B) Second pass
- (C) Pre-pass

(D) Post-pass

Q.11 Who is recognized as the "Father of the Computer" for conceiving the Analytical Engine?

(A) Blaise Pascal

(B) Charles Babbage

(C) Herman Hollerith

(D) John von Neumann

Q.12 John von Neumann's 1945 paper established the foundation for:

(A) Stored-program digital computer architecture

(B) The internet

(C) Quantum computing

(D) Microprocessor lithography

Q.13 Which electronic component defined first-generation computing systems?

(A) Vacuum tubes

(B) Transistors

(C) Integrated circuits

(D) Microprocessors

Q.14 Which of the following was a notable second-generation computer improvement over first-generation systems?

(A) Use of transistors resulting in smaller size, less heat, and greater reliability

(B) Elimination of all programming languages

(C) Introduction of the World Wide Web

(D) Use of mechanical gears

Q.15 Integrated circuits (ICs), combining multiple electronic components on a single chip, marked the:

(A) First generation

(B) Second generation

(C) Third generation

(D) Fifth generation

Q.16 The first single-chip commercial microprocessor, the Intel 4004, launched which computer generation?

(A) Third generation

(B) Fourth generation

(C) Second generation

(D) Fifth generation

Q.17 Current computing research emphasizing parallel multi-core architectures, neural networks, and AI belongs to:

(A) Second generation

(B) Third generation

(C) Fifth generation

(D) First generation

Q.18 The computer characteristic GIGO stands for:

(A) General Input, General Output

(B) Garbage In, Garbage Out

(C) Global Instruction, Global Operation

(D) Guaranteed Input, Guaranteed Output

Q.19 Which CPU unit directs instruction execution and coordinates data flow between memory and the ALU?

(A) Arithmetic Logic Unit

(B) Control Unit (CU)

(C) System Clock

(D) Hard drive

Q.20 Which system bus transmits physical memory addresses from the processor to memory chips?

(A) Data Bus

(B) Address Bus

(C) Control Bus

(D) Power Bus

Q.21 Which of the following is an example of non-volatile primary storage used for startup firmware?

(A) RAM

(B) ROM

(C) L1 Cache

(D) L2 Cache

Q.22 Device drivers are classified as:

(A) System software

(B) Application software

(C) Spreadsheet tools

(D) Optical hardware

Q.23 Which class of computer is specifically designed to support high-volume transaction processing for airlines and banks?

(A) Microcontroller

(B) Mainframe computer

(C) Workstation

(D) Tablet

Q.24 A microcontroller combines which components on a single silicon chip?

(A) Processor, memory, and I/O interfaces

(B) Only a screen and keyboard

(C) Only a cooling fan

(D) A power supply and hard disk

Q.25 A computer designed to perform a single dedicated task, such as engine fuel-injection control, is a:

(A) General-purpose computer

(B) Special-purpose / Embedded computer

(C) Mainframe computer

(D) Supercomputer

Answer Key & Explanations — CA-1 (Set D)

CA-1 Set D — Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	6	B	11	B	16	B	21	B
2	B	7	B	12	A	17	C	22	A
3	B	8	A	13	A	18	B	23	B
4	A	9	A	14	A	19	B	24	A
5	B	10	B	15	C	20	B	25	B

Technical Solutions — CA-1 Set D

- **Q.2 (B):** Compilers translate code prior to execution and catch syntax errors ahead of runtime.
- **Q.6 (B):** The operating system loader allocates memory space and places the compiled executable into RAM.
- **Q.11 (B):** Charles Babbage designed the Analytical Engine, establishing basic computer organization concepts.
- **Q.16 (B):** The Intel 4004 (1971) initiated the 4th generation of microprocessors.
- **Q.24 (A):** A microcontroller is a complete "system on a chip" incorporating CPU, RAM, ROM, and I/O pins.

Chapter 2

Mid-Term Examination Suite

The Mid-Term Examination evaluates students across **Unit 1 (Computer Languages)**, **Unit 2 (Computer Fundamentals)**, and **Unit 3 (Computer Hardware)**. Each paper consists of **30 Multiple-Choice Questions (1 mark each, Total = 30 Marks, Time Allowed = 60 Minutes)**. Complete Answer Keys and technical explanations are appended to each set.

2.1 Mid-Term Examination — Set A

LOVELY PROFESSIONAL UNIVERSITY

Department of Computer Science & Engineering

Mid-Term Examination — SET A

Course Code: CSE111

Title: Orientation to Computing

Time Allowed: 60 Minutes

Max. Marks: 30 Marks

Syllabus Scope: Units 1, 2 & 3 (Languages, Fundamentals & Hardware)

Format: 100% MCQ

Instructions: All 30 questions are compulsory. Each question carries 1 mark. Select the single best answer.

Q.1 Which language is executed directly by the CPU without any intermediate translation?

- (A) Assembly language
- (B) Machine language
- (C) C language
- (D) Python

Q.2 In assembly language, the operation symbol MOV is known as a/an:

- (A) Operand
- (B) Mnemonic
- (C) Register
- (D) Variable

Q.3 A translator that translates and executes high-level source code line-by-line during runtime is a/an:

- (A) Compiler
- (B) Interpreter
- (C) Assembler
- (D) Linker

Q.4 Which phase of compilation converts character streams into tokens such as keywords and identifiers?

- (A) Lexical Analysis
- (B) Syntax Analysis
- (C) Semantic Analysis
- (D) Code Generation

Q.5 What is the primary role of the Linker in the C/C++ compilation pipeline?

- (A) To write source code
- (B) To combine object files with library functions into a final executable
- (C) To load the program into memory
- (D) To execute machine code

Q.6 Which system software loads the binary executable from disk into RAM to start execution?

- (A) Compiler
- (B) Loader
- (C) Assembler
- (D) Editor

Q.7 What does an assembler do during its first pass over assembly source code?

- (A) Emits machine code bytes
- (B) Builds a symbol table recording label names and memory addresses
- (C) Executes instructions
- (D) Formats the output

Q.8 An error that occurs when a program compiles successfully but produces incorrect output is a:

- (A) Syntax error
- (B) Logic error
- (C) Compile-time error
- (D) Linker error

Q.9 What is the correct sequence of the CPU instruction execution cycle?

- (A) Decode → Execute → Fetch → Store
- (B) Fetch → Decode → Execute → Store
- (C) Store → Fetch → Execute → Decode
- (D) Execute → Fetch → Decode → Store

Q.10 Who proposed the stored-program architecture in 1945 where instructions and data share the same memory?

- (A) Blaise Pascal
- (B) John von Neumann
- (C) Charles Babbage
- (D) Herman Hollerith

Q.11 First-generation computers (1940–1956) utilized which primary electronic switching component?

- (A) Transistors
- (B) Vacuum tubes
- (C) Integrated circuits
- (D) Microprocessors

Q.12 Second-generation computers (1956–1963) replaced vacuum tubes with:

- (A) Microprocessors
- (B) Transistors
- (C) Integrated circuits
- (D) Magnetic drums

Q.13 Integrated circuits (ICs) on silicon chips were the defining technology of which computer generation?

- (A) First generation
- (B) Second generation
- (C) Third generation
- (D) Fourth generation

Q.14 Which company launched the world's first commercial single-chip microprocessor (the 4004) in 1971?

- (A) IBM
- (B) Intel
- (C) Apple
- (D) Microsoft

Q.15 Fifth-generation computing primarily emphasizes which of the following technological domains?

- (A) Vacuum tubes
- (B) Artificial Intelligence and massive parallel processing
- (C) Mechanical calculators
- (D) Punched card tabulators

Q.16 The principle that incorrect input data leads to incorrect output information is abbreviated as:

- (A) FIFO
- (B) GIGO
- (C) LIFO
- (D) DMA

Q.17 Which CPU component performs arithmetic calculations and logical comparisons?

- (A) Control Unit
- (B) Arithmetic Logic Unit (ALU)
- (C) Instruction Register
- (D) Program Counter

Q.18 Which system bus is unidirectional, carrying memory addresses from the CPU to memory chips?

- (A) Data Bus
- (B) Address Bus
- (C) Control Bus
- (D) Power Bus

Q.19 Which of the following is volatile primary memory?

- (A) ROM
- (B) Hard disk
- (C) RAM
- (D) SSD

Q.20 A dedicated microcontroller built inside a washing machine or microwave to perform a fixed control task is an:

- (A) Supercomputer
- (B) Embedded computer
- (C) Mainframe
- (D) Workstation

Q.21 Why are GPUs preferred over CPUs for training deep learning neural networks?

- (A) GPUs have higher clock frequencies for single threads
- (B) GPUs have thousands of smaller cores for massive parallel arithmetic
- (C) GPUs eliminate RAM
- (D) GPUs are non-volatile

Q.22 The CPU register that stores the memory address of the next instruction to be fetched is the:

- (A) Instruction Register (IR)
- (B) Program Counter (PC)
- (C) Accumulator
- (D) Stack Pointer

Q.23 The CPU register that holds the instruction currently being executed or decoded is the:

- (A) Program Counter (PC)
- (B) Instruction Register (IR)
- (C) Memory Address Register
- (D) Status Register

Q.24 Which level of CPU cache is typically the fastest and closest to the CPU processing core?

- (A) L3 Cache
- (B) L2 Cache
- (C) L1 Cache
- (D) Virtual memory

Q.25 Dynamic RAM (DRAM) differs from Static RAM (SRAM) because DRAM:

- (A) Is non-volatile
- (B) Requires periodic electrical refreshing
- (C) Is used for CPU cache
- (D) Is made of optical discs

Q.26 Which memory retains its contents permanently even after the computer is powered off?

- (A) DRAM
- (B) SRAM
- (C) ROM
- (D) L1 Cache

Q.27 Solid-State Drives (SSDs) are faster and more shock-resistant than Hard Disk Drives (HDDs) because:

- (A) SSDs use rotating magnetic platters
- (B) SSDs use flash memory with no moving mechanical parts
- (C) SSDs use water-cooling
- (D) SSDs are connected to USB

Q.28 How many bits are in one standard Byte?

- (A) 4
- (B) 8
- (C) 16
- (D) 32

Q.29 A touchscreen display is classified as:

- (A) An input device only
- (B) An output device only
- (C) Both an input and an output device
- (D) A secondary storage medium

Q.30 Which software tool is required to allow the operating system to communicate with a specific peripheral device like a printer?

- (A) Compiler
- (B) Device driver
- (C) Text editor
- (D) Linker

Answer Key & Explanations — Mid-Term (Set A)

Mid-Term Set A — Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	7	B	13	C	19	C	25	B
2	B	8	B	14	B	20	B	26	C
3	B	9	B	15	B	21	B	27	B
4	A	10	B	16	B	22	B	28	B
5	B	11	B	17	B	23	B	29	C
6	B	12	B	18	B	24	C	30	B

Technical Solutions — Mid-Term Set A

- **Q.4 (A):** Lexical analysis scans the source characters and generates tokens for the compiler.
- **Q.10 (B):** John von Neumann's architecture holds programs and data in the same unified memory.
- **Q.21 (B):** GPUs feature thousands of cores executing identical arithmetic instructions on large data arrays in parallel.
- **Q.22 (B):** The Program Counter (PC) stores the address of the next instruction to fetch.
- **Q.27 (B):** SSDs have no mechanical read heads or spinning disks, making them durable and fast.

2.2 Mid-Term Examination — Set B

LOVELY PROFESSIONAL UNIVERSITY

Department of Computer Science & Engineering

Mid-Term Examination — SET B

Course Code: CSE111

Title: Orientation to Computing

Time Allowed: 60 Minutes

Max. Marks: 30 Marks

Syllabus Scope: Units 1, 2 & 3 (Languages, Fundamentals & Hardware)

Format: 100% MCQ

Instructions: All 30 questions are compulsory. Each question carries 1 mark. Select the single best answer.

Q.1 Which of the following programming languages provides machine independence and human readability?

- (A) Machine Language
- (B) High-Level Language
- (C) Microcode
- (D) Hardware Description

Q.2 A program that translates an entire high-level program into machine code prior to execution is a/an:

- (A) Interpreter
- (B) Compiler
- (C) Assembler
- (D) Text Editor

Q.3 In an assembly instruction `MOV R1, #5`, what does `#5` represent?

- (A) An opcode
- (B) An immediate operand value
- (C) A memory address pointer
- (D) A label

Q.4 Checking for type compatibility between variables and operators is performed during:

- (A) Lexical Analysis
- (B) Syntax Analysis
- (C) Semantic Analysis
- (D) Code Generation

Q.5 What does the preprocessor in C/C++ handle before compilation?

- (A) Memory allocation
- (B) Header file inclusions and macro replacements
- (C) Linking object files
- (D) Bytecode generation

Q.6 Which tool resolves external library calls (e.g., `printf`) and merges object files?

- (A) Compiler
- (B) Linker
- (C) Loader
- (D) Debugger

Q.7 What is the primary task of a two-pass assembler during its second pass?

- (A) Building the symbol table
- (B) Generating binary machine instructions
- (C) Checking spelling of comments
- (D) Running the program

Q.8 Testing program behavior with boundary conditions like *hours* = 0 is called:

- (A) Boundary value testing
- (B) Syntax parsing
- (C) Lexical scanning
- (D) Semantic checking

Q.9 Which stage of the CPU instruction cycle determines what operation needs to be performed?

- (A) Fetch
- (B) Decode
- (C) Execute
- (D) Store

Q.10 Who invented the mechanical adding machine known as the Pascaline in 1642?

- (A) Charles Babbage
- (B) Blaise Pascal
- (C) Herman Hollerith
- (D) Alan Turing

Q.11 The Analytical Engine containing concepts of store and mill was designed in the 1830s by:

- (A) Blaise Pascal
- (B) Charles Babbage
- (C) John von Neumann
- (D) Herman Hollerith

Q.12 Herman Hollerith used which technology to automate data compilation for the 1890 US Census?

- (A) Vacuum tubes
- (B) Punched cards
- (C) Transistors
- (D) Integrated circuits

Q.13 First-generation computers (1940–1956) were characterized by:

- (A) Extremely small size and low heat
- (B) Large size, high power consumption, and vacuum tubes
- (C) Silicon microprocessors
- (D) Optical storage

Q.14 Second-generation computers (1956–1963) introduced:

- (A) Integrated circuits
- (B) Transistors and high-level languages like FORTRAN and COBOL
- (C) The internet
- (D) Artificial Intelligence

Q.15 Keyboards, monitors, and multiprogramming operating systems appeared during which generation?

- (A) First generation
- (B) Second generation
- (C) Third generation
- (D) Fifth generation

Q.16 The fourth generation of computers (from 1971) is primarily marked by:

- (A) Vacuum tubes
- (B) Single-chip microprocessors and personal computers
- (C) Relays
- (D) Punched cards

Q.17 Fifth-generation computing focuses on:

- (A) Vacuum tubes
- (B) Artificial Intelligence, parallel processing, and neural networks
- (C) Punched card tabulators
- (D) Analog mechanical gears

Q.18 The computer's ability to perform repetitive operations continuously without fatigue is termed:

- (A) Speed
- (B) Versatility
- (C) Diligence
- (D) Automation

Q.19 Which CPU component performs arithmetic additions and logical comparisons?

- (A) Control Unit
- (B) Arithmetic Logic Unit (ALU)
- (C) Instruction Register
- (D) Address Bus

Q.20 Which system bus transmits physical memory addresses from the CPU to memory chips?

- (A) Data Bus
- (B) Address Bus
- (C) Control Bus
- (D) Serial Bus

Q.21 Which system bus conveys read, write, and interrupt commands between hardware units?

- (A) Data Bus
- (B) Address Bus
- (C) Control Bus
- (D) Power Bus

Q.22 High-performance computers used for weather forecasting, molecular modeling, and physics simulations are:

- (A) Microcontrollers
- (B) Supercomputers
- (C) Workstations
- (D) Embedded systems

Q.23 Large, reliable computers used by banks and government agencies for high-volume transactions are:

- (A) Mainframe computers
- (B) Workstations
- (C) Laptops
- (D) Microcontrollers

Q.24 A dedicated computer built directly into an automobile braking system or microwave oven is a/an:

- (A) Supercomputer
- (B) Embedded computer
- (C) Mainframe
- (D) Server

Q.25 A single-user desktop system with high-end CPU and graphics for CAD and 3D engineering is a:

- (A) Mainframe
- (B) Workstation
- (C) Embedded system
- (D) Supercomputer

Q.26 Which of the following is non-volatile primary memory that stores startup firmware (BIOS/UEFI)?

- (A) RAM
- (B) ROM
- (C) L1 Cache
- (D) L2 Cache

Q.27 Dynamic RAM (DRAM) is used as main system memory because:

- (A) It uses flip-flops and needs no refresh
- (B) It provides high density at moderate cost, though it requires periodic electrical refreshing
- (C) It is non-volatile
- (D) It is made of magnetic tape

Q.28 Static RAM (SRAM) is used in CPU cache memories primarily because:

- (A) It is very inexpensive
- (B) It is faster than DRAM and does not require periodic refreshing
- (C) It has higher capacity than hard drives
- (D) It retains data when unpowered

Q.29 A mechanical Hard Disk Drive (HDD) is more prone to physical shock damage than an SSD because:

- (A) It uses high-voltage lasers
- (B) It contains moving mechanical read/write heads and rotating platters
- (C) It is made of silicon chips
- (D) It has no moving parts

Q.30 An input device that converts physical sound waves into digital electrical signals is a:

- (A) Speaker
- (B) Microphone
- (C) Monitor
- (D) Printer

Answer Key & Explanations — Mid-Term (Set B)

Mid-Term Set B — Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	7	B	13	B	19	B	25	B
2	B	8	A	14	B	20	B	26	B
3	B	9	B	15	C	21	C	27	B
4	C	10	B	16	B	22	B	28	B
5	B	11	B	17	B	23	A	29	B
6	B	12	B	18	C	24	B	30	B

Technical Solutions — Mid-Term Set B

- **Q.4 (C):** Semantic analysis checks for type mismatches, variable declarations, and scope rules.
- **Q.9 (B):** The Decode stage determines the operation and required operands from the fetched opcode.
- **Q.20 (B):** The Address Bus carries the physical memory address from the CPU to memory chips.
- **Q.27 (B):** DRAM stores bits in tiny capacitors that lose charge, requiring dynamic refreshing.
- **Q.28 (B):** SRAM uses transistor flip-flops, making it much faster than DRAM and ideal for CPU cache.

2.3 Mid-Term Examination — Set C

LOVELY PROFESSIONAL UNIVERSITY

Department of Computer Science & Engineering

Mid-Term Examination — SET C

Course Code: CSE111

Title: Orientation to Computing

Time Allowed: 60 Minutes

Max. Marks: 30 Marks

Syllabus Scope: Units 1, 2 & 3 (Languages, Fundamentals & Hardware)

Format: 100% MCQ

Instructions: All 30 questions are compulsory. Each question carries 1 mark. Select the single best answer.

Q.1 In machine language, instructions are represented using:

- (A) Alphabets only
- (B) Binary digits (0 and 1)
- (C) Hexadecimal numbers only
- (D) ASCII text

Q.2 What is the primary function of an Assembler?

- (A) To convert high-level languages into machine code
- (B) To convert assembly language mnemonics into machine code
- (C) To execute code line-by-line
- (D) To edit documents

Q.3 Which of the following is a high-level programming language?

- (A) Machine code
- (B) Java
- (C) Assembly code
- (D) Microcode

Q.4 Which phase of compilation groups source characters into tokens such as keywords and operators?

- (A) Lexical Analysis
- (B) Syntax Analysis
- (C) Semantic Analysis
- (D) Optimization

Q.5 In C/C++, header file expansions (like `#include <stdio.h>`) are handled by the:

- (A) Compiler
- (B) Preprocessor
- (C) Linker
- (D) Loader

Q.6 The software that combines multiple compiled object files with libraries into a single binary executable is the:

- (A) Preprocessor
- (B) Linker
- (C) Loader
- (D) Debugger

Q.7 What does the operating system's loader do?

- (A) Translates code to assembly
- (B) Loads the executable from disk into main memory to run
- (C) Finds syntax errors
- (D) Edits code

Q.8 An error where the program runs but calculates $2 + 2 = 5$ because of a wrong formula is a:

- (A) Syntax error
- (B) Logic error
- (C) Linker error
- (D) Compiler warning

Q.9 What happens during the *Fetch* stage of the CPU instruction cycle?

- (A) The ALU performs addition
- (B) The instruction is retrieved from memory into a CPU register
- (C) The instruction is decoded
- (D) The output is printed

Q.10 A finite, unambiguous sequence of steps designed to solve a computing problem is an:

- (A) Mnemonic
- (B) Algorithm
- (C) Operand
- (D) Opcode

Q.11 Who designed the Analytical Engine in the 1830s containing early concepts of memory and processor?

- (A) Blaise Pascal
- (B) Charles Babbage
- (C) Herman Hollerith
- (D) Alan Turing

Q.12 The stored-program concept was introduced in 1945 by:

- (A) John von Neumann
- (B) Charles Babbage
- (C) Linus Torvalds
- (D) Steve Jobs

Q.13 First-generation computers (1940–1956) were characterized by the use of:

- (A) Transistors
- (B) Vacuum tubes
- (C) Integrated circuits
- (D) Microprocessors

Q.14 Which hardware component replaced vacuum tubes in second-generation computers (1956–1963)?

- (A) Microprocessors
- (B) Transistors
- (C) Integrated circuits
- (D) Magnetic drums

Q.15 Integrated circuits (ICs) on single silicon chips were introduced in the:

- (A) First generation
- (B) Second generation
- (C) Third generation
- (D) Fifth generation

Q.16 The fourth generation of computers (starting in 1971) was defined by the development of:

- (A) Vacuum tubes
- (B) Single-chip microprocessors
- (C) Relays
- (D) Gears

Q.17 Fifth-generation computing focuses on which modern development area?

- (A) Vacuum tubes
- (B) Artificial Intelligence and parallel computing
- (C) Punched cards
- (D) Magnetic core memory

Q.18 The computer characteristic GIGO stands for:

- (A) Guaranteed Input, Guaranteed Output
- (B) Garbage In, Garbage Out
- (C) Global Instruction, General Output
- (D) General Input, Good Output

Q.19 Which CPU unit directs instruction execution and coordinates data flow between memory and the ALU?

- (A) Arithmetic Logic Unit
- (B) Control Unit (CU)
- (C) System Clock
- (D) Hard drive

Q.20 Which system bus transmits physical memory addresses from the processor to memory chips?

- (A) Data Bus
- (B) Address Bus
- (C) Control Bus
- (D) Power Bus

Q.21 Which system bus transfers actual instructions and data words between the CPU and memory?

- (A) Control Bus
- (B) Data Bus
- (C) Address Bus
- (D) Serial Bus

Q.22 High-performance computers used for scientific simulations, weather forecasting, and defense research are:

- (A) Microcontrollers
- (B) Supercomputers
- (C) Workstations
- (D) Embedded systems

Q.23 Large, fault-tolerant computers used by banks and government agencies for high-volume transactions are:

- (A) Personal computers
- (B) Mainframe computers
- (C) Laptops
- (D) Microcontrollers

Q.24 A small computer built directly into an automobile engine or washing machine to perform a dedicated task is a/an:

- (A) Supercomputer
- (B) Embedded computer
- (C) Mainframe
- (D) Server

Q.25 A computer that processes continuous physical variables like voltage or temperature is a/an:

- (A) Digital computer
- (B) Analog computer
- (C) Hybrid computer
- (D) Mainframe

Q.26 Why are GPUs preferred over CPUs for training deep learning models?

- (A) GPUs have higher clock speeds for single threads
- (B) GPUs contain thousands of cores designed for parallel matrix operations
- (C) GPUs eliminate the need for RAM
- (D) GPUs are non-volatile

Q.27 The CPU register that stores the memory address of the next instruction to fetch is the:

- (A) Instruction Register (IR)
- (B) Program Counter (PC)
- (C) Accumulator
- (D) Stack Pointer

Q.28 The CPU register that holds the instruction currently being executed or decoded is the:

- (A) Program Counter (PC)
- (B) Instruction Register (IR)
- (C) Memory Buffer Register
- (D) Status Register

Q.29 Which level of CPU cache is typically the fastest and closest to the processor execution core?

- (A) L3 Cache
- (B) L2 Cache
- (C) L1 Cache
- (D) Virtual memory

Q.30 Which printer technology uses powdered toner and an electrostatic drum to produce rapid documents?

- (A) Inkjet printer
- (B) Laser printer
- (C) Dot matrix printer
- (D) Thermal printer

Answer Key & Explanations — Mid-Term (Set C)

Mid-Term Set C — Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	7	B	13	B	19	B	25	B
2	B	8	B	14	B	20	B	26	B
3	B	9	B	15	C	21	B	27	B
4	A	10	B	16	B	22	B	28	B
5	B	11	B	17	B	23	B	29	C
6	B	12	A	18	B	24	B	30	B

Technical Solutions — Mid-Term Set C

- **Q.2 (B):** Assemblers translate assembly mnemonics into machine code on a 1-to-1 basis.
- **Q.9 (B):** The Fetch stage reads the instruction from memory into the CPU using the Program Counter.
- **Q.26 (B):** GPUs are massively parallel, processing thousands of data elements simultaneously, making them ideal for neural networks.
- **Q.27 (B):** The PC (Program Counter) points to the next instruction in sequence.
- **Q.30 (B):** Laser printers fuse powdered toner with heat and pressure onto paper.

2.4 Mid-Term Examination — Set D

LOVELY PROFESSIONAL UNIVERSITY

Department of Computer Science & Engineering

Mid-Term Examination — SET D

Course Code: CSE111

Title: Orientation to Computing

Time Allowed: 60 Minutes

Max. Marks: 30 Marks

Syllabus Scope: Units 1, 2 & 3 (Languages, Fundamentals & Hardware)

Format: 100% MCQ

Instructions: All 30 questions are compulsory. Each question carries 1 mark. Select the single best answer.

Q.1 Which language level provides maximum hardware control and execution speed but lacks portability?

- (A) High-Level Language
- (B) Machine Language
- (C) Python
- (D) Visual Basic

Q.2 In an assembly instruction `ADD R1, R2`, what is the operation word `ADD`?

- (A) An operand
- (B) A mnemonic opcode
- (C) A label
- (D) A directive

Q.3 A compiler differs from an interpreter because a compiler:

- (A) Translates line-by-line at runtime
- (B) Translates the entire program into machine/object code before execution
- (C) Requires no CPU
- (D) Executes only in the cloud

Q.4 The compilation stage that checks whether keywords and operators form valid grammatical statements is:

- (A) Lexical Analysis
- (B) Syntax Analysis
- (C) Semantic Analysis
- (D) Loading

Q.5 A missing semicolon in a C++ source file causes which type of error?

- (A) Compile-time syntax error
- (B) Runtime error
- (C) Logic error
- (D) Linker warning

Q.6 What is the primary role of the preprocessor in the C language?

- (A) Translates C code into assembly
- (B) Expands macros and header inclusions before compilation
- (C) Loads binary into RAM
- (D) Executes code

Q.7 What does an assembler do in its first pass?

- (A) Emits binary machine code
- (B) Records symbol names and memory addresses in a symbol table
- (C) Links object files
- (D) Tests program logic

Q.8 Which step of the program development cycle involves formalizing the problem requirements, inputs, and outputs?

- (A) Problem Definition
- (B) Coding
- (C) Testing
- (D) Maintenance

Q.9 An informal description of an algorithm written in plain English mixed with programming-style logic is:

- (A) Pseudocode
- (B) Machine code
- (C) Bytecode
- (D) Microcode

Q.10 Who is celebrated as the "Father of the Computer" for proposing the Analytical Engine?

- (A) Blaise Pascal
- (B) Charles Babbage
- (C) Herman Hollerith
- (D) John von Neumann

Q.11 The 1945 stored-program architecture where code and data reside in common memory was formulated by:

- (A) John von Neumann
- (B) Charles Babbage
- (C) Alan Turing
- (D) Linus Torvalds

Q.12 First-generation computers (1940–1956) were characterized by which physical switching devices?

- (A) Transistors
- (B) Vacuum tubes
- (C) Integrated circuits
- (D) Microprocessors

Q.13 Second-generation computers (1956–1963) introduced which electronic component, reducing heat and physical size?

- (A) Vacuum tubes
- (B) Transistors
- (C) Integrated circuits
- (D) Microprocessors

Q.14 Integrated circuits (ICs) combining multiple components on a single silicon chip marked which generation?

- (A) First generation
- (B) Second generation
- (C) Third generation
- (D) Fifth generation

Q.15 The Intel 4004 (1971) was the world's first commercial:

- (A) Single-chip microprocessor
- (B) Mainframe computer
- (C) Operating system
- (D) Laser printer

Q.16 Fifth-generation computing research focuses heavily on:

- (A) Relay switches
- (B) Artificial Intelligence and parallel computing
- (C) Vacuum tubes

(D) Mechanical punch cards

Q.17 The computer characteristic GIGO reminds us that:

- (A) Fast computers never make mistakes
- (B) Output correctness depends strictly on input accuracy
- (C) Hard drives lose data
- (D) Compilers fix logic errors

Q.18 Which CPU component performs mathematical calculations (+, −) and logical comparisons (AND, OR)?

- (A) Control Unit
- (B) Arithmetic Logic Unit (ALU)
- (C) Instruction Register
- (D) Address Bus

Q.19 Which system bus is unidirectional and carries physical memory addresses from the CPU to memory chips?

- (A) Data Bus
- (B) Address Bus
- (C) Control Bus
- (D) Power Bus

Q.20 Which system bus carries command signals like Memory Read, Memory Write, and Interrupts?

- (A) Data Bus
- (B) Address Bus
- (C) Control Bus
- (D) Expansion Bus

Q.21 Which memory type provides ultra-fast temporary storage located directly inside the CPU chip?

- (A) Registers
- (B) Hard disk
- (C) Flash drive
- (D) Optical disc

Q.22 Why is primary RAM called "volatile" memory?

- (A) It gets physically hot
- (B) It loses its stored data as soon as power is disconnected
- (C) It cannot be overwritten

(D) It uses spinning disks

Q.23 Which type of software includes operating systems like Windows and Linux?

(A) Application software

(B) System software

(C) Utility spreadsheet

(D) Firmware patch

Q.24 Computers designed to process continuous physical measurements (like temperature or pressure) are:

(A) Digital computers

(B) Analog computers

(C) Hybrid computers

(D) Mainframes

Q.25 A computer built directly into an anti-lock braking system (ABS) to perform dedicated real-time control is a/an:

(A) Supercomputer

(B) Embedded computer

(C) Mainframe

(D) Workstation

Q.26 Why are GPUs preferred over CPUs for training deep learning models?

(A) GPUs have higher clock speeds for single threads

(B) GPUs contain thousands of cores designed for parallel matrix operations

(C) GPUs eliminate the need for RAM

(D) GPUs are non-volatile

Q.27 The CPU register that stores the memory address of the next instruction to fetch is the:

(A) Instruction Register (IR)

(B) Program Counter (PC)

(C) Accumulator

(D) Stack Pointer

Q.28 Dynamic RAM (DRAM) is used as main system memory because:

(A) It uses flip-flops and needs no refresh

(B) It provides high density at moderate cost, though it requires periodic electrical refreshing

(C) It is non-volatile

(D) It is made of magnetic tape

Q.29 Solid-State Drives (SSDs) are faster and more durable than Hard Disk Drives (HDDs) because:

(A) SSDs have spinning platters

(B) SSDs use flash memory with no moving mechanical parts

(C) SSDs use high-power lasers

(D) SSDs are water-cooled

Q.30 Which input peripheral converts printed text on paper into editable digital characters using OCR?

(A) Microphone

(B) Scanner

(C) Barcode reader

(D) Touchpad

Answer Key & Explanations — Mid-Term (Set D)

Mid-Term Set D — Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	7	B	13	B	19	B	25	B
2	B	8	A	14	C	20	C	26	B
3	B	9	A	15	A	21	A	27	B
4	B	10	B	16	B	22	B	28	B
5	A	11	A	17	B	23	B	29	B
6	B	12	B	18	B	24	B	30	B

Technical Solutions — Mid-Term Set D

- **Q.3 (B):** Compilers translate the whole program before execution, allowing fast, native execution.
- **Q.5 (A):** Syntax errors are grammar violations detected during the compile phase.
- **Q.18 (B):** The Arithmetic Logic Unit handles all mathematical and Boolean operations.
- **Q.27 (B):** The Program Counter (PC) register holds the address of the next instruction to fetch.
- **Q.29 (B):** SSDs have zero moving mechanical parts, providing silent, fast, and shock-resistant storage.

Chapter 3

Continuous Assessment 2 (CA-2) Suite

Continuous Assessment 2 evaluates students on **Unit 4 (Number Systems)** and **Unit 5 (Version Control)**. Each paper contains **25 Multiple-Choice Questions (1 mark each, Total = 25 Marks, Time Allowed = 45 Minutes)**. Complete Answer Keys and step-by-step conversion solutions are provided.

3.1 Continuous Assessment 2 — Set A

LOVELY PROFESSIONAL UNIVERSITY**Department of Computer Science & Engineering****Continuous Assessment 2 — SET A****Course Code:** CSE111**Title:** Orientation to Computing**Time Allowed:** 45 Minutes**Max. Marks:** 25 Marks**Syllabus Scope:** Unit 4 (Number Systems) & Unit 5 (Version Control)**Format:** 100% MCQ

Instructions: All 25 questions are compulsory. Each question carries 1 mark. Select the single best answer.

Q.1 What is the base (radix) of the Binary number system?

- (A) 2
- (B) 8
- (C) 10
- (D) 16

Q.2 What is the base (radix) of the Octal number system?

- (A) 2
- (B) 8
- (C) 10
- (D) 16

Q.3 What is the base (radix) of the Hexadecimal number system?

- (A) 8
- (B) 10
- (C) 12
- (D) 16

Q.4 Which of the following is an INVALID octal number?

- (A) 107
- (B) 543
- (C) 782
- (D) 615

Q.5 In the hexadecimal system, what decimal value is represented by the letter 'A'?

- (A) 10
- (B) 11
- (C) 12
- (D) 15

Q.6 In the hexadecimal system, what decimal value is represented by the letter 'F'?

- (A) 10
- (B) 12
- (C) 14
- (D) 15

Q.7 What is the decimal equivalent of the binary number $(1011)_2$?

- (A) 9
- (B) 11
- (C) 13
- (D) 15

Q.8 How many binary bits correspond directly to a single Octal digit?

- (A) 2 bits
- (B) 3 bits
- (C) 4 bits
- (D) 8 bits

Q.9 How many binary bits correspond directly to a single Hexadecimal digit?

- (A) 2 bits
- (B) 3 bits
- (C) 4 bits
- (D) 8 bits

Q.10 What is the octal equivalent of the binary number $(110101)_2$?

- (A) $(55)_8$
- (B) $(65)_8$
- (C) $(35)_8$
- (D) $(71)_8$

Q.11 What is the hexadecimal equivalent of the binary number $(10101111)_2$?

- (A) $(AF)_{16}$

(B) $(FA)_{16}$

(C) $(BF)_{16}$

(D) $(AE)_{16}$

Q.12 What is the decimal value of the binary fraction $(0.1)_2$?

(A) 0.1

(B) 0.5

(C) 0.25

(D) 0.125

Q.13 In a positional number, what is the leftmost non-zero digit called?

(A) Least Significant Digit (LSD)

(B) Most Significant Digit (MSD)

(C) Radix digit

(D) Fractional bit

Q.14 Who designed and created the Git version control system in 2005?

(A) Dennis Ritchie

(B) Linus Torvalds

(C) James Gosling

(D) Guido van Rossum

Q.15 What type of version control architecture is used by Git?

(A) Centralized only

(B) Distributed version control system

(C) Peer-to-peer torrent system

(D) Manual logging

Q.16 What is GitHub?

(A) A command-line text editor

(B) A cloud-based hosting and collaboration service for Git repositories

(C) A programming language compiler

(D) A CPU component

Q.17 Which hidden directory is created by Git inside a project folder to store tracking metadata?

(A) `.github`

(B) `.git`

(C) `.version`

(D) `.repo`

Q.18 Which command initializes a new local Git repository in the current directory?

(A) `git start`

(B) `git init`

(C) `git create`

(D) `git new`

Q.19 Which Git command checks the status of modified, staged, and untracked files?

(A) `git status`

(B) `git check`

(C) `git log`

(D) `git inspect`

Q.20 In Git's three-stage architecture, what is the intermediate area where changes are prepared before being committed?

(A) Working Directory

(B) Staging Area (Index)

(C) Remote Server

(D) Trash Bin

Q.21 Which command is used to move changes from the working directory to the Git staging area?

(A) `git commit`

(B) `git add <filename>`

(C) `git save`

(D) `git stage`

Q.22 Which command permanently records the staged snapshot into the local repository history?

(A) `git save`

(B) `git commit -m "message"`

(C) `git push`

(D) `git freeze`

Q.23 What is the purpose of the `.gitignore` file?

(A) To list files that Git should intentionally ignore and not track

(B) To delete old commits

- (C) To store passwords
- (D) To speed up downloads

Q.24 In Git, what is a branch?

- (A) A separate hard drive
- (B) A lightweight movable pointer to a specific commit
- (C) A deleted repository
- (D) An untracked image file

Q.25 Which command uploads local branch commits to a remote repository like GitHub?

- (A) `git pull`
- (B) `git push`
- (C) `git fetch`
- (D) `git commit`

Answer Key & Explanations — CA-2 (Set A)

CA-2 Set A — Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	6	D	11	A	16	B	21	B
2	B	7	B	12	B	17	B	22	B
3	D	8	B	13	B	18	B	23	A
4	C	9	C	14	B	19	A	24	B
5	A	10	B	15	B	20	B	25	B

Technical Solutions — CA-2 Set A

- **Q.4 (C):** The digit '8' is invalid in Octal (base 8 allows digits 0 to 7).
- **Q.7 (B):** $1011_2 = 1(8) + 0(4) + 1(2) + 1(1) = 11_{10}$.
- **Q.10 (B):** 110101_2 grouped in 3-bit sets: $\underbrace{110}_6 \underbrace{101}_5 = (65)_8$.
- **Q.11 (A):** 10101111_2 grouped in 4-bit sets: $\underbrace{1010}_A \underbrace{1111}_F = (AF)_{16}$.
- **Q.14 (B):** Linus Torvalds created Git in 2005 to manage Linux kernel source code.
- **Q.21 (B):** `git add` moves files into the staging area (index).
- **Q.25 (B):** `git push` transmits commits from the local branch to the remote repository.

3.2 Continuous Assessment 2 — Set B

LOVELY PROFESSIONAL UNIVERSITY

Department of Computer Science & Engineering

Continuous Assessment 2 — SET B

Course Code: CSE111

Title: Orientation to Computing

Time Allowed: 45 Minutes

Max. Marks: 25 Marks

Syllabus Scope: Unit 4 (Number Systems) & Unit 5 (Version Control)

Format: 100% MCQ

Instructions: All 25 questions are compulsory. Each question carries 1 mark. Select the single best answer.

Q.1 Which digits are valid in the Binary number system?

- (A) 0 and 1 only
- (B) 0 through 7
- (C) 0 through 9
- (D) 0 through 9 and A through F

Q.2 In the hexadecimal number system, what decimal value does the letter 'C' represent?

- (A) 10
- (B) 11
- (C) 12
- (D) 13

Q.3 In the hexadecimal number system, what decimal value does the letter 'E' represent?

- (A) 12
- (B) 13
- (C) 14
- (D) 15

Q.4 What is the decimal equivalent of the binary number $(1111)_2$?

- (A) 14
- (B) 15
- (C) 16
- (D) 17

Q.5 What is the decimal equivalent of the binary number $(1001)_2$?

- (A) 7
- (B) 8
- (C) 9
- (D) 10

Q.6 What is the binary representation of the decimal integer 13_{10} ?

- (A) $(1101)_2$
- (B) $(1011)_2$
- (C) $(1110)_2$
- (D) $(1001)_2$

Q.7 What is the binary equivalent of the octal number $(73)_8$?

- (A) $(111011)_2$
- (B) $(110011)_2$
- (C) $(111010)_2$
- (D) $(101011)_2$

Q.8 What is the hexadecimal equivalent of the binary number $(11001010)_2$?

- (A) $(CA)_{16}$
- (B) $(AC)_{16}$
- (C) $(BA)_{16}$
- (D) $(C9)_{16}$

Q.9 In binary addition, what is the result of $1 + 1$?

- (A) 0 with carry 1 (10_2)
- (B) 1 with no carry
- (C) 11_2
- (D) 0 with no carry

Q.10 What is the decimal value of the binary fraction $(0.01)_2$?

- (A) 0.5
- (B) 0.25
- (C) 0.125
- (D) 0.01

Q.11 In positional notation, what is the separator between the integer and fractional parts?

- (A) Radix point

- (B) Hyphen
- (C) Modulo
- (D) Colon

Q.12 What is the decimal equivalent of the octal number $(25)_8$?

- (A) 17
- (B) 21
- (C) 25
- (D) 29

Q.13 What is the decimal equivalent of the hexadecimal number $(1A)_{16}$?

- (A) 24
- (B) 26
- (C) 28
- (D) 30

Q.14 Which command stages all modified and untracked files in the current folder?

- (A) `git add .`
- (B) `git commit -a`
- (C) `git push -all`
- (D) `git save .`

Q.15 What does the `-m` flag stand for in `git commit -m "msg"`?

- (A) Message
- (B) Module
- (C) Modify
- (D) Merge

Q.16 A file that has been added to the staging area is in which Git state?

- (A) Untracked
- (B) Staged
- (C) Pushed
- (D) Ignored

Q.17 Which command displays the past commits condensed into a single line per commit?

- (A) `git log -oneline`
- (B) `git history -short`

(C) `git log -brief`

(D) `git summary`

Q.18 What is the default name for the primary initial branch in modern Git repositories?

(A) `root`

(B) `main` or `master`

(C) `origin`

(D) `head`

Q.19 Which command creates a new branch named `feature-nav` and switches to it?

(A) `git switch -c feature-nav`

(B) `git branch feature-nav`

(C) `git make feature-nav`

(D) `git checkout feature-nav`

Q.20 Which command sets your global author name in Git configuration?

(A) `git config -global user.name "Your Name"`

(B) `git set name "Your Name"`

(C) `git user.name = "Your Name"`

(D) `git init -name "Your Name"`

Q.21 Which command downloads changes from a remote repository and immediately merges them into the current branch?

(A) `git fetch`

(B) `git pull`

(C) `git push`

(D) `git clone`

Q.22 Which command creates an identical local copy of a remote GitHub repository?

(A) `git copy URL`

(B) `git clone URL`

(C) `git download URL`

(D) `git import URL`

Q.23 Does saving a file inside an editor like VS Code automatically create a Git commit?

(A) Yes, always

(B) No, saving on disk does not stage or commit in Git

- (C) Only on macOS
- (D) Only for HTML files

Q.24 On GitHub, what feature allows developers to propose changes from their branch and request code review?

- (A) Issue tracker
- (B) Pull Request (PR)
- (C) Git bash
- (D) Fork bomb

Q.25 Can Git be used to track changes locally on a laptop without an active internet connection?

- (A) No, internet is required
- (B) Yes, all core Git operations work offline locally
- (C) Only for Python files
- (D) Only on Linux

Answer Key & Explanations — CA-2 (Set B)

CA-2 Set B — Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	6	A	11	A	16	B	21	B
2	C	7	A	12	B	17	A	22	B
3	C	8	A	13	B	18	B	23	B
4	B	9	A	14	A	19	A	24	B
5	C	10	B	15	A	20	A	25	B

Technical Solutions — CA-2 Set B

- **Q.2 (C):** In Hexadecimal, $A = 10, B = 11, C = 12$.
- **Q.6 (A):** $13_{10} = 8 + 4 + 0 + 1 = 1101_2$.
- **Q.10 (B):** $0.01_2 = 0 \times 2^{-1} + 1 \times 2^{-2} = 0.25_{10}$.
- **Q.13 (B):** $(1A)_{16} = 1(16) + 10 = 26_{10}$.
- **Q.21 (B):** `git pull` fetches remote commits and merges them into the working branch.
- **Q.25 (B):** Git is a distributed VCS and stores history locally in the `.git` folder, working completely offline.

3.3 Continuous Assessment 2 — Set C

LOVELY PROFESSIONAL UNIVERSITY**Department of Computer Science & Engineering****Continuous Assessment 2 — SET C****Course Code:** CSE111**Title:** Orientation to Computing**Time Allowed:** 45 Minutes**Max. Marks:** 25 Marks**Syllabus Scope:** Unit 4 (Number Systems) & Unit 5 (Version Control)**Format:** 100% MCQ

Instructions: All 25 questions are compulsory. Each question carries 1 mark. Select the single best answer.

Q.1 In the hexadecimal system, what decimal value does the letter 'B' represent?

- (A) 10
- (B) 11
- (C) 12
- (D) 13

Q.2 What is the decimal equivalent of the binary number $(1010)_2$?

- (A) 8
- (B) 10
- (C) 12
- (D) 14

Q.3 What is the decimal equivalent of the binary number $(1100)_2$?

- (A) 10
- (B) 12
- (C) 14
- (D) 16

Q.4 What is the binary representation of the decimal integer 8_{10} ?

- (A) $(1000)_2$
- (B) $(0100)_2$
- (C) $(1100)_2$
- (D) $(1111)_2$

Q.5 What is the octal equivalent of the binary number $(101111)_2$?

- (A) $(57)_8$
- (B) $(47)_8$
- (C) $(67)_8$
- (D) $(55)_8$

Q.6 What is the binary equivalent of the hexadecimal number $(F5)_{16}$?

- (A) $(11110101)_2$
- (B) $(11100101)_2$
- (C) $(11110110)_2$
- (D) $(10100101)_2$

Q.7 In binary addition, what is the result of $1 + 1 + 1$?

- (A) 10_2
- (B) 11_2
- (C) 01_2
- (D) 100_2

Q.8 What is the decimal value of the binary fraction $(0.101)_2$?

- (A) 0.5
- (B) 0.625
- (C) 0.75
- (D) 0.125

Q.9 In a positional number, what is the rightmost digit called?

- (A) Most Significant Digit (MSD)
- (B) Least Significant Digit (LSD)
- (C) Base exponent
- (D) Radix root

Q.10 When converting a decimal integer to binary by repeated division, remainders are read:

- (A) From first remainder to last
- (B) From last remainder (bottom) to first remainder (top)
- (C) In random order
- (D) From smallest to largest

Q.11 What is the decimal equivalent of the octal number $(70)_8$?

- (A) 56

(B) 64

(C) 70

(D) 49

Q.12 What is the decimal equivalent of the hexadecimal number $(20)_{16}$?

(A) 20

(B) 32

(C) 40

(D) 16

Q.13 Why does the decimal fraction 0.1_{10} cause small rounding errors in standard binary arithmetic?

(A) Because 0.1_{10} is an infinitely repeating binary fraction

(B) Because the CPU cannot process decimals

(C) Because binary is base 8

(D) Because 0.1 is an integer

Q.14 Which command removes a file from the staging area while leaving it safe in your working directory?

(A) `git rm -f`

(B) `git restore -staged <filename>`

(C) `git delete`

(D) `git wipe`

Q.15 Which command displays commit history in reverse chronological order?

(A) `git history`

(B) `git log`

(C) `git show-all`

(D) `git list`

Q.16 In the command `git push -u origin feature1`, what does `origin` refer to?

(A) The author's user name

(B) The short name for the remote repository

(C) The local branch

(D) The commit hash

Q.17 Which command lists all local branches in the current repository?

(A) `git branch`

- (B) `git list -branches`
- (C) `git status`
- (D) `git show`

Q.18 Which command sets your global commit email address in Git?

- (A) `git email "your@email.com"`
- (B) `git config -global user.email "your@email.com"`
- (C) `git config set email "your@email.com"`
- (D) `git register -email`

Q.19 Which command downloads remote repository history without automatically merging it into your working branch?

- (A) `git fetch`
- (B) `git pull`
- (C) `git push`
- (D) `git commit`

Q.20 What is a unique 40-character hexadecimal string identifying a Git commit called?

- (A) A commit hash / SHA-1 checksum
- (B) A PIN number
- (C) A file extension
- (D) A branch label

Q.21 A GitHub repository that can only be viewed and accessed by authorized collaborators is a:

- (A) Public repository
- (B) Private repository
- (C) Open-source repository
- (D) Forked repository

Q.22 What does `git status` report when all working files have been staged and committed?

- (A) "Fatal error"
- (B) "Nothing to commit, working tree clean"
- (C) "Staging overflow"
- (D) "Remote disconnected"

Q.23 What happens if you delete the hidden `.git` folder from a project directory?

- (A) All source code is deleted

- (B) Git tracking history is removed, turning the folder into an unversioned directory
- (C) The computer operating system crashes
- (D) GitHub repository is deleted

Q.24 What is the primary function of the `README.md` file in a GitHub repository?

- (A) To store passwords
- (B) To provide project overview, setup instructions, and documentation
- (C) To compile C++ code
- (D) To hold binary images

Q.25 Which file specifies files and patterns that Git should deliberately ignore and not track?

- (A) `config.sys`
- (B) `.gitignore`
- (C) `git.config`
- (D) `README.md`

Answer Key & Explanations — CA-2 (Set C)

CA-2 Set C — Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	6	A	11	A	16	B	21	B
2	B	7	B	12	B	17	A	22	B
3	B	8	B	13	A	18	B	23	B
4	A	9	B	14	B	19	A	24	B
5	A	10	B	15	B	20	A	25	B

Technical Solutions — CA-2 Set C

- **Q.5 (A):** 101111_2 in 3-bit groups: $\underbrace{101}_5 \underbrace{111}_7 = (57)_8$.
- **Q.8 (B):** $0.101_2 = 1(0.5) + 0(0.25) + 1(0.125) = 0.625_{10}$.
- **Q.11 (A):** $(70)_8 = 7(8^1) + 0(8^0) = 56_{10}$.
- **Q.12 (B):** $(20)_{16} = 2(16^1) + 0(16^0) = 32_{10}$.
- **Q.19 (A):** `git fetch` downloads remote commits without merging, unlike `git pull` which merges automatically.

3.4 Continuous Assessment 2 — Set D

LOVELY PROFESSIONAL UNIVERSITY**Department of Computer Science & Engineering****Continuous Assessment 2 — SET D****Course Code:** CSE111**Title:** Orientation to Computing**Time Allowed:** 45 Minutes**Max. Marks:** 25 Marks**Syllabus Scope:** Unit 4 (Number Systems) & Unit 5 (Version Control)**Format:** 100% MCQ

Instructions: All 25 questions are compulsory. Each question carries 1 mark. Select the single best answer.

Q.1 In hexadecimal notation, what decimal value does the letter 'D' represent?

- (A) 11
- (B) 12
- (C) 13
- (D) 14

Q.2 What is the decimal equivalent of the binary number $(1110)_2$?

- (A) 12
- (B) 13
- (C) 14
- (D) 15

Q.3 What is the decimal equivalent of the binary number $(1000)_2$?

- (A) 4
- (B) 6
- (C) 8
- (D) 16

Q.4 What is the binary representation of the decimal integer 11_{10} ?

- (A) $(1011)_2$
- (B) $(1101)_2$
- (C) $(1110)_2$
- (D) $(1001)_2$

Q.5 What is the octal equivalent of the binary number $(111010)_2$?

- (A) $(72)_8$
- (B) $(62)_8$
- (C) $(52)_8$
- (D) $(74)_8$

Q.6 What is the hexadecimal equivalent of the binary number $(11110000)_2$?

- (A) $(F0)_{16}$
- (B) $(0F)_{16}$
- (C) $(E0)_{16}$
- (D) $(FF)_{16}$

Q.7 What is the value of $(101)_2 + (010)_2$?

- (A) $(111)_2$
- (B) $(110)_2$
- (C) $(100)_2$
- (D) $(1000)_2$

Q.8 What is the decimal value of the binary fraction $(0.11)_2$?

- (A) 0.5
- (B) 0.75
- (C) 0.25
- (D) 0.125

Q.9 In a positional numeral system, how does place value change as you move one position to the left of the radix point?

- (A) Multiplies by the base (b)
- (B) Divides by the base (b)
- (C) Remains unchanged
- (D) Squares the digit

Q.10 What is the decimal equivalent of the octal number $(33)_8$?

- (A) 24
- (B) 27
- (C) 30
- (D) 33

Q.11 What is the decimal equivalent of the hexadecimal number $(2B)_{16}$?

- (A) 41
- (B) 43
- (C) 45
- (D) 47

Q.12 In web design, what does the hexadecimal color code #0000FF represent?

- (A) Pure Red
- (B) Pure Green
- (C) Pure Blue
- (D) Pure Black

Q.13 A byte contains 8 bits. What is the maximum decimal value that can be stored in an unsigned 8-bit byte?

- (A) 127
- (B) 128
- (C) 255
- (D) 256

Q.14 Which command initializes a brand new Git repository inside the current project folder?

- (A) `git start`
- (B) `git init`
- (C) `git create`
- (D) `git new`

Q.15 Which command stages all modified, deleted, and untracked files in the current folder?

- (A) `git add .`
- (B) `git commit -a`
- (C) `git push -all`
- (D) `git save .`

Q.16 Which command commits staged files with a clear descriptive message?

- (A) `git commit -m "Your message"`
- (B) `git save -m "Your message"`
- (C) `git push -m "Your message"`
- (D) `git record "Your message"`

Q.17 In the output of `git branch`, what does an asterisk (*) beside a branch name mean?

- (A) The branch is empty
- (B) The branch is the currently checked-out branch
- (C) The branch is locked
- (D) The branch was deleted

Q.18 Which command switches to an existing branch named `testing`?

- (A) `git switch testing`
- (B) `git move testing`
- (C) `git branch testing`
- (D) `git jump testing`

Q.19 Which command sets your global author name in Git configuration?

- (A) `git config -global user.name "Your Name"`
- (B) `git set name "Your Name"`
- (C) `git user.name = "Your Name"`
- (D) `git init -name "Your Name"`

Q.20 Which command uploads local branch commits to GitHub?

- (A) `git pull`
- (B) `git push`
- (C) `git fetch`
- (D) `git commit`

Q.21 Which command downloads remote repository commits and integrates them into your current branch?

- (A) `git fetch`
- (B) `git pull`
- (C) `git push`
- (D) `git clone`

Q.22 Which command makes an exact local copy of a remote GitHub repository?

- (A) `git copy URL`
- (B) `git clone URL`
- (C) `git download URL`
- (D) `git import URL`

Q.23 Does creating a commit locally automatically publish it to GitHub?

- (A) Yes, immediately
- (B) No, commits stay local until pushed with `git push`
- (C) Only on Windows
- (D) Only for text files

Q.24 On GitHub, what feature allows team members to propose, review, and discuss changes before merging?

- (A) Pull Request (PR)
- (B) Issue tracker
- (C) Terminal
- (D) Compiler

Q.25 Which file lists temporary and sensitive files that Git should deliberately omit from tracking?

- (A) `.gitignore`
- (B) `README.md`
- (C) `config.sys`
- (D) `git.log`

Answer Key & Explanations — CA-2 (Set D)

CA-2 Set D — Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	6	A	11	B	16	A	21	B
2	C	7	A	12	C	17	B	22	B
3	C	8	B	13	C	18	A	23	B
4	A	9	A	14	B	19	A	24	A
5	A	10	B	15	A	20	B	25	A

Technical Solutions — CA-2 Set D

- **Q.1 (C):** In Hexadecimal, A = 10, B = 11, C = 12, D = 13.
- **Q.5 (A):** 111010_2 grouped in threes: $\underbrace{111}_7 \underbrace{010}_2 = (72)_8$.
- **Q.8 (B):** $0.11_2 = 1(0.5) + 1(0.25) = 0.75_{10}$.
- **Q.11 (B):** $(2B)_{16} = 2(16) + 11 = 32 + 11 = 43_{10}$.
- **Q.13 (C):** $2^8 - 1 = 255$ is the maximum unsigned integer value for 8 bits.
- **Q.23 (B):** `git commit` only captures snapshots into the local machine's `.git` folder.

Chapter 4

End-Term Examination Suite (Comprehensive)

The End-Term Examination is the comprehensive objective test covering all six modules of **CSE111: Orientation to Computing**:

- **Unit 1:** Computer Languages (Machine, Assembly, HLL, Compilers, Interpreters, Development Cycle)
- **Unit 2:** Computer Fundamentals (Von Neumann Architecture, Generations 1–5, Functional Units, System Buses)
- **Unit 3:** Computer Hardware (CPU vs. GPU, Peripherals, Memory Hierarchy, DRAM vs. SRAM, ROM, SSDs)
- **Unit 4:** Number Systems (Binary, Octal, Decimal, Hexadecimal, Direct Grouping Conversions)
- **Unit 5:** Version Control (Git Distributed Model, Three-Tree Architecture, Branching, GitHub)
- **Unit 6:** Modern AI Trends and Tools (AI/ML/DL, Generative AI, LLMs, Research Tools, Prompt Engineering)

Each comprehensive paper consists of **30 MCQs (1 mark each, Total = 30 Marks, Time Allowed = 90 Minutes)**. Complete Answer Keys and technical rationales follow each set.

4.1 End-Term Examination — Set A

LOVELY PROFESSIONAL UNIVERSITY

Department of Computer Science & Engineering

End-Term Examination (Comprehensive) — SET A

Course Code: CSE111	Title: Orientation to Computing	Time Allowed: 90 Minutes
Max. Marks: 30 Marks	Syllabus Scope: Comprehensive Coverage: All Units (Units 1 to 6)	Format: 100% MCQ

Instructions: All 30 questions are compulsory. Each question carries 1 mark. Select the single best answer.

Q.1 Which of the following programming languages is directly executed by the CPU without translation?

- (A) Assembly language
- (B) Machine language
- (C) Python
- (D) C++

Q.2 A software tool that converts high-level source code into machine code statement-by-statement at runtime is a/an:

- (A) Compiler
- (B) Interpreter
- (C) Assembler
- (D) Linker

Q.3 The software responsible for combining compiled object files and libraries into a runnable executable is the:

- (A) Loader
- (B) Linker
- (C) Compiler
- (D) Editor

Q.4 The stored-program architecture where instructions and data reside in common memory was formulated in 1945 by:

- (A) Blaise Pascal
- (B) John von Neumann
- (C) Charles Babbage

(D) Alan Turing

Q.5 What electronic switching component was used in first-generation computers (1940–1956)?

(A) Transistors

(B) Vacuum tubes

(C) Integrated circuits

(D) Microprocessors

Q.6 The fourth generation of computers (starting in 1971) was defined by the development of:

(A) Vacuum tubes

(B) Microprocessors

(C) Magnetic drums

(D) Relays

Q.7 Which CPU component performs arithmetic calculations and logical comparisons?

(A) Control Unit

(B) Arithmetic Logic Unit (ALU)

(C) Program Counter

(D) Address Bus

Q.8 Which system bus is unidirectional and transmits physical memory addresses from the CPU to memory chips?

(A) Data Bus

(B) Address Bus

(C) Control Bus

(D) Power Bus

Q.9 Why are GPUs preferred over CPUs for training deep learning models?

(A) GPUs have higher clock speeds for single threads

(B) GPUs contain thousands of smaller cores for parallel matrix arithmetic

(C) GPUs eliminate RAM

(D) GPUs are non-volatile

Q.10 The CPU register that stores the memory address of the next instruction to fetch is the:

(A) Instruction Register (IR)

(B) Program Counter (PC)

(C) Accumulator

(D) Stack Pointer

Q.11 Dynamic RAM (DRAM) differs from Static RAM (SRAM) because DRAM:

- (A) Is non-volatile
- (B) Requires periodic electrical refreshing to retain data
- (C) Is faster than SRAM
- (D) Is used in CPU cache

Q.12 Solid-State Drives (SSDs) are faster and more shock-resistant than Hard Disk Drives (HDDs) because:

- (A) SSDs use spinning platters
- (B) SSDs use flash memory chips with no moving parts
- (C) SSDs use lasers
- (D) SSDs are connected to USB

Q.13 What is the base (radix) of the Hexadecimal number system?

- (A) 8
- (B) 10
- (C) 12
- (D) 16

Q.14 In the hexadecimal number system, what decimal value does the letter 'C' represent?

- (A) 10
- (B) 11
- (C) 12
- (D) 13

Q.15 What is the decimal equivalent of the binary number $(1011)_2$?

- (A) 9
- (B) 11
- (C) 13
- (D) 15

Q.16 How many binary bits correspond directly to a single Octal digit?

- (A) 2 bits
- (B) 3 bits
- (C) 4 bits

(D) 8 bits

Q.17 What is the octal equivalent of the binary number $(110101)_2$?

(A) $(55)_8$

(B) $(65)_8$

(C) $(35)_8$

(D) $(71)_8$

Q.18 What is the hexadecimal equivalent of the binary number $(10101111)_2$?

(A) $(AF)_{16}$

(B) $(FA)_{16}$

(C) $(BF)_{16}$

(D) $(AE)_{16}$

Q.19 Who created the Git version control system in 2005?

(A) Dennis Ritchie

(B) Linus Torvalds

(C) James Gosling

(D) Guido van Rossum

Q.20 What kind of version control system is Git?

(A) Centralized

(B) Distributed

(C) Single-user

(D) Manual

Q.21 Which command initializes a new local Git repository in the current folder?

(A) `git start`

(B) `git init`

(C) `git create`

(D) `git new`

Q.22 Which command stages modified and untracked files in the current directory?

(A) `git add .`

(B) `git commit -a`

(C) `git push`

(D) `git save`

Q.23 Which command commits staged files with a clear descriptive message?

- (A) `git save -m "msg"`
- (B) `git commit -m "msg"`
- (C) `git push -m "msg"`
- (D) `git record "msg"`

Q.24 Which command uploads local branch commits to GitHub?

- (A) `git pull`
- (B) `git push`
- (C) `git fetch`
- (D) `git clone`

Q.25 The term "Artificial Intelligence" was formally coined at which historic 1956 academic conference?

- (A) The Turing Conference
- (B) The Dartmouth Summer Research Project
- (C) IEEE World Congress
- (D) Bell Labs Conference

Q.26 All existing commercial AI assistants (like Siri, ChatGPT, and Google Gemini) are classified as:

- (A) Artificial General Intelligence (AGI)
- (B) Narrow AI
- (C) Super AI
- (D) Self-conscious AI

Q.27 When a generative language model produces confident-sounding statements that are factually false, it is called:

- (A) Overclocking
- (B) Hallucination
- (C) Paging
- (D) Thrashing

Q.28 What does "multimodal AI" (such as Google Gemini) mean?

- (A) It runs on multiple monitors
- (B) It can understand and process text, images, and audio simultaneously
- (C) It has multiple power cables
- (D) It requires no internet

Q.29 Which AI research assistant grounds its answers strictly in the user's uploaded documents and notes?

- (A) Midjourney
- (B) NotebookLM
- (C) DALL-E
- (D) Synthesia

Q.30 In prompt engineering, assigning a persona such as "Act as a computer science professor" establishes the:

- (A) Task
- (B) Role
- (C) Constraint
- (D) Format

Answer Key & Explanations — End-Term (Set A)

End-Term Set A — Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	7	B	13	D	19	B	25	B
2	B	8	B	14	C	20	B	26	B
3	B	9	B	15	B	21	B	27	B
4	B	10	B	16	B	22	A	28	B
5	B	11	B	17	B	23	B	29	B
6	B	12	B	18	A	24	B	30	B

Technical Solutions — End-Term Set A

- **Q.2 (B):** Interpreters translate and execute statement-by-statement during runtime.
- **Q.9 (B):** GPUs contain thousands of lightweight parallel cores, making them far more efficient for tensor math than CPUs.
- **Q.14 (C):** In Hexadecimal, C = 12.
- **Q.17 (B):** 110101₂ in 3-bit groups: $\underbrace{110}_6 \underbrace{101}_5 = (65)_8$.
- **Q.27 (B):** Hallucination is the fluent generation of factually incorrect information by LLMs.
- **Q.29 (B):** Google's NotebookLM is grounded strictly in the user's provided notes and documents.

4.2 End-Term Examination — Set B

LOVELY PROFESSIONAL UNIVERSITY

Department of Computer Science & Engineering

End-Term Examination (Comprehensive) — SET B

Course Code: CSE111	Title: Orientation to Computing	Time Allowed: 90 Minutes
Max. Marks: 30 Marks	Syllabus Scope: Comprehensive Coverage: All Units (Units 1 to 6)	Format: 100% MCQ

Instructions: All 30 questions are compulsory. Each question carries 1 mark. Select the single best answer.

Q.1 In assembly language, readable operation abbreviations like ADD and MOV are:

- (A) Operands
- (B) Mnemonics
- (C) Macros
- (D) Pointers

Q.2 A compiler translates a high-level program:

- (A) Statement-by-statement during runtime
- (B) Entirely into machine/object code prior to execution
- (C) Without using a CPU
- (D) Exclusively in the cloud

Q.3 The operating system loader is responsible for:

- (A) Checking syntax errors
- (B) Placing the executable image into physical RAM to initiate execution
- (C) Compiling Java source code
- (D) Formatting hard disks

Q.4 First-generation computers (1940–1956) were characterized by:

- (A) Silicon microprocessors
- (B) Vacuum tubes, high power consumption, and substantial heat
- (C) Transistors
- (D) Integrated circuits

Q.5 Second-generation computers (1956–1963) replaced vacuum tubes with:

- (A) Microprocessors
- (B) Transistors
- (C) Integrated circuits
- (D) Magnetic drums

Q.6 Integrated circuits (ICs) on silicon chips defined which computer generation?

- (A) First generation
- (B) Second generation
- (C) Third generation
- (D) Fifth generation

Q.7 The world's first single-chip commercial microprocessor, the Intel 4004, launched in:

- (A) 1956
- (B) 1964
- (C) 1971
- (D) 1981

Q.8 Which CPU component directs instruction execution and coordinates data flow between memory and the ALU?

- (A) Arithmetic Logic Unit
- (B) Control Unit (CU)
- (C) System Clock
- (D) Hard drive

Q.9 Which system bus transfers actual instructions and data words between the CPU and memory?

- (A) Address Bus
- (B) Control Bus
- (C) Data Bus
- (D) Serial Bus

Q.10 A small computer built directly into an automobile engine or washing machine to perform a dedicated task is a/an:

- (A) Supercomputer
- (B) Embedded computer
- (C) Mainframe
- (D) Workstation

Q.11 The CPU register that holds the instruction currently being executed or decoded is the:

- (A) Program Counter (PC)
- (B) Instruction Register (IR)
- (C) Memory Buffer Register
- (D) Status Register

Q.12 Which level of CPU cache is typically the fastest and closest to the execution core?

- (A) L3 Cache
- (B) L2 Cache
- (C) L1 Cache
- (D) Virtual memory

Q.13 What is the primary difference between Dynamic RAM (DRAM) and Static RAM (SRAM)?

- (A) DRAM is non-volatile while SRAM is volatile
- (B) DRAM requires periodic electrical refreshing while SRAM does not
- (C) DRAM is faster than SRAM
- (D) SRAM is used for hard disk storage

Q.14 Which memory type retains its contents permanently even when the computer is powered off?

- (A) DRAM
- (B) SRAM
- (C) ROM
- (D) L2 Cache

Q.15 In the hexadecimal number system, what decimal value does the letter 'B' represent?

- (A) 10
- (B) 11
- (C) 12
- (D) 13

Q.16 What is the decimal equivalent of the binary number $(1111)_2$?

- (A) 14
- (B) 15
- (C) 16
- (D) 17

Q.17 How many binary bits correspond directly to a single Hexadecimal digit?

- (A) 2 bits

- (B) 3 bits
- (C) 4 bits
- (D) 8 bits

Q.18 What is the binary equivalent of the octal number $(73)_8$?

- (A) $(111011)_2$
- (B) $(110011)_2$
- (C) $(111010)_2$
- (D) $(101011)_2$

Q.19 What is the decimal value of the binary fraction $(0.01)_2$?

- (A) 0.5
- (B) 0.25
- (C) 0.125
- (D) 0.01

Q.20 In a positional number, what is the rightmost digit called?

- (A) Most Significant Digit (MSD)
- (B) Least Significant Digit (LSD)
- (C) Base exponent
- (D) Radix root

Q.21 Which hidden folder is created by Git to store repository metadata and history?

- (A) `.github`
- (B) `.git`
- (C) `.repo`
- (D) `.config`

Q.22 In Git's three-stage architecture, what is the intermediate area where changes are prepared before committing?

- (A) Working Directory
- (B) Staging Area (Index)
- (C) Remote Repository
- (D) Trash Bin

Q.23 Which command moves modified files to the Git staging area?

- (A) `git commit`

- (B) `git add <file>`
- (C) `git push`
- (D) `git save`

Q.24 What is the purpose of the `.gitignore` file?

- (A) To list files that Git should intentionally ignore and not track
- (B) To delete old commits
- (C) To store passwords
- (D) To speed up downloads

Q.25 A machine learning model adjusting its internal weights using a large training dataset is undergoing:

- (A) Inference
- (B) Training
- (C) Tokenization
- (D) Formatting

Q.26 Applying an already-trained model to produce predictions or answer user questions is called:

- (A) Training
- (B) Inference
- (C) Compiling
- (D) Formatting

Q.27 Which conversational AI tool is developed by OpenAI?

- (A) ChatGPT
- (B) Gemini
- (C) NotebookLM
- (D) Midjourney

Q.28 Which of the following is a popular text-to-image AI generation tool?

- (A) DALL-E
- (B) Perplexity AI
- (C) Git Bash
- (D) Linux

Q.29 Which AI research assistant provides web search answers accompanied by clickable source citations?

- (A) Perplexity AI

- (B) Midjourney
- (C) Runway
- (D) Synthesia

Q.30 In prompt engineering, constraints such as "Limit your answer to 150 words" represent:

- (A) The Role
- (B) The Constraint
- (C) The Model
- (D) The Token

Answer Key & Explanations — End-Term (Set B)

End-Term Set B — Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	7	C	13	B	19	B	25	B
2	B	8	B	14	C	20	B	26	B
3	B	9	C	15	B	21	B	27	A
4	B	10	B	16	B	22	B	28	A
5	B	11	B	17	C	23	B	29	A
6	C	12	C	18	A	24	A	30	B

Technical Solutions — End-Term Set B

- **Q.13 (B):** DRAM stores bits in tiny capacitors that lose charge and must be constantly refreshed.
- **Q.15 (B):** In Hexadecimal, A = 10, B = 11.
- **Q.18 (A):** $(73)_8 = \underbrace{111}_7 \underbrace{011}_3 = 111011_2$.
- **Q.26 (B):** Inference refers to running a trained model to generate answers for new inputs.
- **Q.29 (A):** Perplexity AI combines search indexing with direct citations for verification.

4.3 End-Term Examination — Set C

LOVELY PROFESSIONAL UNIVERSITY**Department of Computer Science & Engineering****End-Term Examination (Comprehensive) — SET C****Course Code:** CSE111**Title:** Orientation to Computing**Time Allowed:** 90 Minutes**Max. Marks:** 30 Marks**Syllabus Scope:** Comprehensive Coverage: All
Units (Units 1 to 6)**Format:** 100% MCQ

Instructions: All 30 questions are compulsory. Each question carries 1 mark. Select the single best answer.

Q.1 Which programming language category uses mnemonics like SUB and JMP on a 1-to-1 machine basis?

- (A) Assembly Language
- (B) Machine Language
- (C) Python
- (D) SQL

Q.2 A missing semicolon in a C++ program causes a:

- (A) Compile-time syntax error
- (B) Runtime error
- (C) Logic error
- (D) Linker error

Q.3 In C/C++, header file inclusions like `#include <stdio.h>` are handled before compilation by the:

- (A) Compiler
- (B) Preprocessor
- (C) Linker
- (D) Loader

Q.4 Who invented the mechanical adding machine known as the Pascaline in 1642?

- (A) Charles Babbage
- (B) Blaise Pascal
- (C) Herman Hollerith
- (D) Alan Turing

Q.5 Herman Hollerith used punched cards to automate data compilation for the:

- (A) 1890 US Census
- (B) Apollo 11 mission
- (C) 1945 codebreaking
- (D) 1971 microprocessor launch

Q.6 Which high-level programming languages were widely popularized during the second generation (1956–1963)?

- (A) Python and Swift
- (B) FORTRAN and COBOL
- (C) HTML and SQL
- (D) C# and Rust

Q.7 The computer characteristic stating that "incorrect input data inevitably produces incorrect output" is:

- (A) FIFO
- (B) GIGO
- (C) LIFO
- (D) DMA

Q.8 Which CPU component performs arithmetic additions and logical comparisons?

- (A) Control Unit
- (B) Arithmetic Logic Unit (ALU)
- (C) Instruction Register
- (D) Address Bus

Q.9 Which system bus is unidirectional and transmits physical memory addresses from the CPU to memory?

- (A) Data Bus
- (B) Address Bus
- (C) Control Bus
- (D) Serial Bus

Q.10 A single-user desktop system with high-end CPU and graphics used for 3D modeling and CAD is a:

- (A) Mainframe
- (B) Workstation
- (C) Embedded system

(D) Supercomputer

Q.11 Why is RAM considered "volatile" memory?

- (A) It gets physically hot
- (B) It loses stored contents as soon as power is disconnected
- (C) It cannot be rewritten
- (D) It uses spinning magnetic platters

Q.12 Which type of memory is commonly used in CPU cache because of its high speed?

- (A) DRAM
- (B) SRAM
- (C) EEPROM
- (D) Magnetic Tape

Q.13 A mechanical Hard Disk Drive (HDD) stores data magnetically on:

- (A) Flash memory cells
- (B) Rotating circular platters with moving read/write heads
- (C) Optical reflective pits
- (D) Punched cards

Q.14 An input device that converts physical sound waves into digital electrical signals is a:

- (A) Speaker
- (B) Microphone
- (C) Monitor
- (D) Printer

Q.15 In hexadecimal notation, what decimal value is represented by the letter 'D'?

- (A) 11
- (B) 12
- (C) 13
- (D) 14

Q.16 What is the decimal equivalent of the binary number $(1001)_2$?

- (A) 7
- (B) 8
- (C) 9
- (D) 10

Q.17 What is the binary representation of the decimal integer 8_{10} ?

- (A) $(1000)_2$
- (B) $(0100)_2$
- (C) $(1100)_2$
- (D) $(1111)_2$

Q.18 What is the hexadecimal equivalent of the binary number $(11001010)_2$?

- (A) $(CA)_{16}$
- (B) $(AC)_{16}$
- (C) $(BA)_{16}$
- (D) $(C9)_{16}$

Q.19 In binary addition, what is the result of $1 + 1$?

- (A) 0 with carry 1 (10_2)
- (B) 1 with no carry
- (C) 11_2
- (D) 0 with no carry

Q.20 What is the decimal value of the binary fraction $(0.101)_2$?

- (A) 0.5
- (B) 0.625
- (C) 0.75
- (D) 0.125

Q.21 Which command is used to initialize a new local Git repository in the current working directory?

- (A) `git start`
- (B) `git init`
- (C) `git create`
- (D) `git new`

Q.22 Which command stages all modified, deleted, and untracked files in the current folder?

- (A) `git add .`
- (B) `git commit -a`
- (C) `git push -all`
- (D) `git save .`

Q.23 In Git, what is a branch?

- (A) A separate hard drive
- (B) A lightweight movable pointer to a specific commit
- (C) A deleted repository
- (D) An untracked image file

Q.24 Which command sets your global author name in Git configuration?

- (A) `git config -global user.name "Your Name"`
- (B) `git set name "Your Name"`
- (C) `git user.name = "Your Name"`
- (D) `git init -name "Your Name"`

Q.25 A subset of Machine Learning that uses multi-layered artificial neural networks is called:

- (A) Deep Learning
- (B) Microcoding
- (C) Direct Memory Access
- (D) Hardware Emulation

Q.26 In natural language processing with AI models, what is a "token"?

- (A) A physical coin
- (B) A piece of text such as a word, sub-word, or punctuation mark
- (C) A secure password
- (D) A network cable

Q.27 Which text-to-image generator is developed by Adobe with a focus on commercially safe design?

- (A) Midjourney
- (B) Adobe Firefly
- (C) ChatGPT
- (D) NotebookLM

Q.28 Which AI writing assistant is specifically tailored to assist researchers with academic drafting and citations?

- (A) JenniAI
- (B) Adobe Firefly
- (C) Midjourney
- (D) Runway

Q.29 In prompt engineering, assigning a persona such as `"Act as a beginner-level tutor"` sets the:

- (A) Role
- (B) Task
- (C) Constraint
- (D) Format

Q.30 Why should students never enter private passwords or confidential student records into public AI tools?

- (A) Because the AI will stop working
- (B) To protect privacy and prevent data from being used in model retraining
- (C) Because inputs can only be 10 characters
- (D) Because AI cannot process names

Answer Key & Explanations — End-Term (Set C)

End-Term Set C — Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	7	B	13	B	19	A	25	A
2	A	8	B	14	B	20	B	26	B
3	B	9	B	15	C	21	B	27	B
4	B	10	B	16	C	22	A	28	A
5	A	11	B	17	A	23	B	29	A
6	B	12	B	18	A	24	A	30	B

Technical Solutions — End-Term Set C

- **Q.12 (B):** SRAM uses flip-flops, providing ultra-low access latency, which makes it the standard choice for CPU cache.
- **Q.15 (C):** In Hexadecimal, $A = 10, B = 11, C = 12, D = 13$.
- **Q.18 (A):** 11001010_2 in 4-bit sets: $\underbrace{1100}_C \underbrace{1010}_A = (CA)_{16}$.
- **Q.20 (B):** $0.101_2 = 1(0.5) + 0(0.25) + 1(0.125) = 0.625_{10}$.
- **Q.25 (A):** Deep learning is defined by the use of multi-layered deep artificial neural networks.

4.4 End-Term Examination — Set D

LOVELY PROFESSIONAL UNIVERSITY

Department of Computer Science & Engineering

End-Term Examination (Comprehensive) — SET D

Course Code: CSE111	Title: Orientation to Computing	Time Allowed: 90 Minutes
Max. Marks: 30 Marks	Syllabus Scope: Comprehensive Coverage: All Units (Units 1 to 6)	Format: 100% MCQ

Instructions: All 30 questions are compulsory. Each question carries 1 mark. Select the single best answer.

Q.1 Which translator translates an entire program into machine code beforehand, reporting errors prior to runtime?

- (A) Interpreter
- (B) Compiler
- (C) Text editor
- (D) Loader

Q.2 A program development step that formalizes the problem inputs, outputs, and constraints before algorithm design is:

- (A) Testing
- (B) Problem Definition
- (C) Maintenance
- (D) Optimization

Q.3 The first mechanical computer to incorporate basic concepts of input, memory, and arithmetic was Babbage's:

- (A) Analytical Engine
- (B) Pascaline
- (C) Abacus
- (D) ENIAC

Q.4 The Intel 4004 (1971) was historically important because it was the first commercial:

- (A) Single-chip microprocessor
- (B) Hard disk drive
- (C) Color monitor
- (D) Compiler

Q.5 Which generation of computers introduced integrated circuits (ICs) on single silicon chips?

- (A) First generation
- (B) Second generation
- (C) Third generation
- (D) Fifth generation

Q.6 Which CPU component performs mathematical additions (+, −) and comparisons?

- (A) Control Unit
- (B) Arithmetic Logic Unit (ALU)
- (C) Instruction Register
- (D) Address Bus

Q.7 Which system bus transmits physical memory addresses from the CPU to memory chips?

- (A) Data Bus
- (B) Address Bus
- (C) Control Bus
- (D) Power Bus

Q.8 Which memory type provides ultra-fast temporary storage located directly inside the CPU chip?

- (A) Registers
- (B) Hard disk
- (C) Flash drive
- (D) Optical disc

Q.9 Why is primary RAM called "volatile" memory?

- (A) It gets physically hot
- (B) It loses its stored data as soon as power is disconnected
- (C) It cannot be overwritten
- (D) It uses spinning disks

Q.10 Why are GPUs preferred over CPUs for training deep learning models?

- (A) GPUs have higher clock speeds for single threads
- (B) GPUs contain thousands of cores designed for parallel matrix operations
- (C) GPUs eliminate the need for RAM
- (D) GPUs are non-volatile

Q.11 The CPU register that stores the memory address of the next instruction to fetch is the:

- (A) Instruction Register (IR)
- (B) Program Counter (PC)
- (C) Accumulator
- (D) Stack Pointer

Q.12 Solid-State Drives (SSDs) are faster and more durable than Hard Disk Drives (HDDs) because:

- (A) SSDs have spinning platters
- (B) SSDs use flash memory with no moving mechanical parts
- (C) SSDs use high-power lasers
- (D) SSDs are water-cooled

Q.13 In the hexadecimal number system, what decimal value is represented by the letter 'E'?

- (A) 12
- (B) 13
- (C) 14
- (D) 15

Q.14 What is the decimal equivalent of the binary number $(1110)_2$?

- (A) 12
- (B) 13
- (C) 14
- (D) 15

Q.15 What is the octal equivalent of the binary number $(111010)_2$?

- (A) $(72)_8$
- (B) $(62)_8$
- (C) $(52)_8$
- (D) $(74)_8$

Q.16 What is the decimal equivalent of the octal number $(33)_8$?

- (A) 24
- (B) 27
- (C) 30
- (D) 33

Q.17 What is the decimal equivalent of the hexadecimal number $(2B)_{16}$?

- (A) 41

(B) 43

(C) 45

(D) 47

Q.18 What is the decimal value of the binary fraction $(0.11)_2$?

(A) 0.5

(B) 0.75

(C) 0.25

(D) 0.125

Q.19 How many bits are in one standard byte?

(A) 4

(B) 8

(C) 16

(D) 32

Q.20 Which command initializes a new local Git repository in the current folder?

(A) `git start`

(B) `git init`

(C) `git create`

(D) `git new`

Q.21 Which command stages all modified, deleted, and untracked files in the current directory?

(A) `git add .`

(B) `git commit -a`

(C) `git push -all`

(D) `git save .`

Q.22 Which command creates a new branch named `feature-cart` and switches to it?

(A) `git switch -c feature-cart`

(B) `git make branch feature-cart`

(C) `git branch feature-cart`

(D) `git move feature-cart`

Q.23 Which command uploads local commits to a remote repository like GitHub?

(A) `git pull`

(B) `git push`

(C) `git fetch`

(D) `git commit`

Q.24 On GitHub, what feature allows developers to propose changes and request team code review?

(A) Pull Request (PR)

(B) Issue tracker

(C) Terminal

(D) Compiler

Q.25 Which file specifies temporary and sensitive files that Git should deliberately omit from tracking?

(A) `.gitignore`

(B) `README.md`

(C) `config.sys`

(D) `git.log`

Q.26 The term "Artificial Intelligence" was formally coined in 1956 at the:

(A) Turing Symposium

(B) Dartmouth Summer Research Project

(C) IEEE Convention

(D) Bell Labs Conference

Q.27 When an AI model produces confident, fluent statements that are factually false, it is called:

(A) Overclocking

(B) Hallucination

(C) Paging

(D) Thrashing

Q.28 Google Gemini is distinguished by being a natively:

(A) Audio-only model

(B) Multimodal model

(C) Analog calculator

(D) Operating system

Q.29 Which AI research assistant answers questions grounded strictly in the user's uploaded documents?

(A) Midjourney

(B) NotebookLM

- (C) DALL-E
- (D) Synthesia

Q.30 In prompt engineering, instructions like "Limit your answer to 100 words in 3 bullet points" represent:

- (A) The Role
- (B) The Constraints and Output Format
- (C) The Model Name
- (D) The Token

Answer Key & Explanations — End-Term (Set D)

End-Term Set D — Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	7	B	13	C	19	B	25	A
2	B	8	A	14	C	20	B	26	B
3	A	9	B	15	A	21	A	27	B
4	A	10	B	16	B	22	A	28	B
5	C	11	B	17	B	23	B	29	B
6	B	12	B	18	B	24	A	30	B

Technical Solutions — End-Term Set D

- **Q.1 (B):** Compilers translate the whole program before execution and detect syntax errors early.
- **Q.13 (C):** In Hexadecimal, A = 10, B = 11, C = 12, D = 13, E = 14.
- **Q.14 (C):** $1110_2 = 8 + 4 + 2 + 0 = 14_{10}$.
- **Q.15 (A):** 111010_2 in 3-bit groups: $\underbrace{111}_7 \underbrace{010}_2 = (72)_8$.
- **Q.17 (B):** $(2B)_{16} = 2(16) + 11 = 43_{10}$.
- **Q.27 (B):** Hallucination is when an LLM fabricates false facts fluently.
- **Q.28 (B):** Multimodal models like Gemini natively process and understand text, images, and audio.